

14

Multiple Regression

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Multiple R^2 versus Unique Effects

In previous chapters, the association between years of education completed and annual earnings was examined in a number of ways. Based on the Pearson correlation, it was found that there was a positive association between years of education completed and annual earnings equal to $r = .337$, which corresponded exactly to the results reported by Zagorsky (2007). Thus, 11.4% of the variance in earnings was accounted for by education. Based on a bivariate regression, it was found that for every additional year of education completed, one would expect, on average, a person to earn \$2,777 more annually. However, it was also found that a regression equation based solely on number of years of education completed would yield, on average, prediction errors equal to a whopping \$24,086!¹ A natural question to ask is whether it would be possible to increase the predictive accuracy of the regression analysis by including a second independent variable to the regression equation. For example, Zagorsky (2007) also found that IQ was related positively to annual earnings ($r = .31$; $r^2 = .096$). Thus, perhaps years of education completed and IQ, in combination, may yield a model R^2 value greater than that based on years of education completed alone (i.e., model $R^2 = .114$).

In the context of multiple regression, model R is also known as multiple R^2 , because it is based on two or more predictors. In addition to a multiple R^2 , multiple regression yields beta weights unique to each predictor included in the model. The word ‘unique’, in this context, refers to a unique contribution to the equation; one that is independent of the effects of any other variable included in the model. IQ and years of education completed was reported by Zagorsky (2007) to be correlated at $r = .57$. Thus, higher levels of intelligence were associated with higher levels of education completed. The fact that IQ and education were correlated positively with earnings, and IQ and education were correlated with each other, makes it difficult to estimate the amount of variance in earnings that can be accounted for by IQ and education combined. That is, it is not simply a process of adding the two r^2 values together, i.e., $.114 + .100 = .214$. Instead, some of the predictive capacity of years of education completed and IQ would be expected to be overlapping/redundant. Thus, the two main tasks of multiple regression are to maximally predict the dependent variable with a regression equation (model R^2), as well as estimate the unique contributions of each predictor variable to the regression equation.

Essentially, multiple regression combines the attributes of regression and semi-partial correlation (type A). Semi-partial correlation is relevant to multiple regression, because only the unique effects associated with the independent variables are included in the multiple regression equation. This is really important to understand. Multiple regression helps decompose the common and unique predictive effects of two or more independent variables

¹ $65.99 \times 365 = 24,086$

on a dependent variable. In some cases, one might have 5 or more hypothesized predictors of a dependent variable. Multiple regression will help determine which of the 5 hypothesized predictors contribute *unique* predictive capacity in predicting the dependent variable. In the years of education completed, IQ, and annual earnings example, a multiple regression will help determine whether years of education completed and IQ can both predict annual earnings statistically significantly and uniquely. If that sounds like a type A semi-partial correlation to you, you're on the right track. The main difference is that multiple regression builds a regression equation that yields **beta weights** to maximize the prediction of the dependent variable (i.e., maximize multiple R^2). By contrast, multiple regression is not directly concerned with semi-partial correlations. So, multiple regression beta weights and semi-partial correlations (type A) have similarities, but they are also somewhat different, as I do describe in further detail below.

Multiple Regression Equation

Mathematically, the outcome of conducting a multiple regression is a regression equation that predicts maximally the dependent variable. To this effect, matrix algebra is used to estimate the terms of the regression equation: one intercept and a beta weight for each of the predictors included in the model. To round out the equation, an error term is also included, as the vast majority of regression equations are associated with a certain amount of error (i.e., they do not predict the dependent variable perfectly). Thus, the multiple regression equation may be specified as follows:

$$\bar{Y} = a + b_1X_1 + b_2X_2 + \dots + b_nX_n + \varepsilon \quad (1)$$

Where $\bar{Y}$ = the predicted value of the dependent variable, a = the intercept, b_1 = the beta weight associated with the first independent variable, b_2 = the beta weight associated with the second independent variable, b_n = the beta weight associated with the n th independent variable, and ε = error. The X_1 , X_2 , X_n values correspond to specified values associated with the independent variables (i.e., particular raw scores). Recall that the purpose of a multiple regression equation is to predict a particular case's value on Y . Once the multiple regression equation has been solved, each case's predicted value on Y can be estimated based on their scores on the independent variables and the application of the regression equation.

Consider the example of years of education completed, IQ, and annual earnings. Suppose someone had 14 years of education completed and an IQ of 112, what would I predict their annual earnings to be? Suppose someone had 8 years of education completed and an IQ of 79, what would I predict their annual earnings to be? I can't answer this until I conduct a multiple regression, because I need an intercept, and two beta weights: one for years of education completed and one for IQ. Before I conduct the multiple regression analysis, I'd like to point out the two different types of beta weights that can be estimated.

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Standardized Versus Unstandardized Beta Weights

There are two types of beta weights estimated in a multiple regression: standardized and unstandardized. Standardized beta weights in multiple regression are associated with the same interpretation as a standardized beta weight in a bivariate regression: a one standard deviation increase in the value of X is associated with a beta-weight percentage of a standard deviation change in the value of Y . Whether the change in Y is an increase or decrease will depend on whether the beta weight is positive or negative in direction. Also, whether the change in Y is large or small will depend on the magnitude of the beta weight. This interpretation of a standardized beta weight is the same as that described in the bivariate regression case. However, there is one very important difference to keep in mind: the beta weights associated with a multiple regression represent unique effects. Recall that in the bivariate case, the standardized beta weight was equal to the Pearson correlation. In practice, this no longer holds true in the multiple regression case. Almost invariably, a variable's beta weight is smaller than its corresponding Pearson correlation. The beta weight is smaller, because it represents only the unique effects on the dependent variable not shared with the other independent variables. So, in the years of education, IQ and annual earnings example, the years of education and IQ beta weights will definitely be smaller than their corresponding Pearson correlations, because education and IQ are correlated with each other positively.

Unstandardized beta weights in multiple regression have the same interpretation as that described in the bivariate regression case, with, again, the important difference that they represent the unique contributions to the regression equation. Typically, in the behavioral sciences, only the standardized beta weights are interpreted and reported. The reason is that many variables in research, particularly the behavioral sciences, do not have a naturally interpretable scale. However, as I will show below, in this example, it makes a lot of sense to report both the unstandardized and standardized beta weights, because years of education, IQ, and annual earnings are measured on a scale that many people understand. One final aspect of multiple regression I would like to describe, before I conduct the analysis, is multiple R^2 .

Multiple R (and Multiple R^2)

In bivariate regression, the Pearson r and the model R are always the same values, because there is only one independent variable in the analysis. However, in multiple regression, model R will always be larger than the Pearson r between any particular independent variable and the dependent variable. The reason is that each independent variable will be expected to contribute some unique variance in predicting the dependent variable. That unique variance may or may not be statistically significant, however, the model

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R will necessarily be larger, at least numerically. In the context of this example, recall that the model R associated with the bivariate regression with education as the independent variable and earnings as the dependent variable was estimated at .337 (as described in the Bivariate Regression chapter). One of the questions posed by conducting a multiple regression is whether adding IQ to the regression equation will increase model R in a statistically significant manner. Next, I conduct the multiple regression analysis in SPSS.

Multiple Regression – Method Enter: SPSS

There are several different methods of multiple regression that one can conduct, including stepwise, hierarchical, and enter, for example. The first type of multiple regression demonstrated in this chapter is known as method ‘enter’, because all of the independent variables are forced into the regression equation, irrespective of whether they are found to be statistically significant contributors to the regression equation or not. In order to demonstrate a multiple regression (enter) analysis in SPSS, I have simulated data to correspond very closely to the results reported by Zygorisky (2007) ([Data File: education earnings N 250](#)). I have simulated the data to correspond to a sample size of 250 (rather than 5,000+) to make things a little more interesting with respect to p -values.² In order to conduct a multiple regression in SPSS, the ‘Linear’ utility can be used (**Watch Video 14.1: [Multiple Regression \(Method Enter\) in SPSS](#)**). The first table outputted by SPSS is entitled ‘Descriptive Statistics’. It provides information on the variable means, standard deviations, and the sample size.

Descriptive Statistics			
	Mean	Std. Deviation	N
earnings	43314.0323	43123.5794	250
education	13.1480	2.66731	250
IQ	99.98	14.933	250

In order to get additional important descriptive statistics, such as skew and kurtosis values, you will need to use the Frequency utility in SPSS which will yield the SPSS table entitled ‘Statistics’ (see Chapter 2).

The main element of the table to point out is that the skew associated with the earnings variable is both positive and very large (3.45), as was observed when the analysis was conducted in the bivariate regression example. For the purposes of learning, I will carry on with the analysis as if the data were sufficiently normally distributed. I will then follow-up with the bootstrapped version of the analysis, which does not assume normally distributed data.

² Note that the sample size used in the correlation and bivariate regression chapters was $N = 40$. Thus, the results reported in this chapter differ slightly to the results reported in chapter 12.

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Statistics		education	earnings	IQ
N	Valid	250	250	250
	Missing	0	0	0
Mean		13.1480	43314.0323	99.98
Median		13.0000	34067.5456	99.00
Std. Deviation		2.66731	43123.5794	14.933
Skewness		.097	3.447	.212
Std. Error of Skewness		.154	.154	.154
Kurtosis		.027	19.820	-.446
Std. Error of Kurtosis		.307	.307	.307
Minimum		5.00	.00	69
Maximum		21.00	382000.00	143

The next table outputted by SPSS is the correlation matrix table entitled 'Correlations'.

Correlations		earnings	education	IQ
Pearson Correlation	earnings	1.000	.332	.307
	education	.332	1.000	.572
	IQ	.307	.572	1.000
Sig. (1-tailed)	earnings	.	.000	.000
	education	.000	.	.000
	IQ	.000	.000	.
N	earnings	250	250	250
	education	250	250	250
	IQ	250	250	250

It can be observed in the SPSS table entitled 'Correlations' that there were positive correlations between all three of the variables included in the analysis. In particular, both education ($r = .332$) and IQ ($r = .307$) correlated positively with earnings. Furthermore, education and IQ were correlated positively with each other ($r = .572$).

The next table outputted by SPSS is entitled 'Variables Entered/Removed'. It includes the names of the variables included in the analysis. Additionally, it specifies the type of method that was used, which, in this case, was 'Enter'. The 'Variables Removed' column only comes into play when one uses multiple regression methods such as 'stepwise', for example, rather than method enter.

Variables Entered/Removed ^a			
Model	Variables Entered	Variables Removed	Method
1	IQ, education ^b	.	Enter

a. Dependent Variable: earnings

b. All requested variables entered.

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Next up is the 'Model Summary' table which provides information relevant to multiple R .

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.361 ^a	.130	.123	40375.3677

a. Predictors: (Constant), IQ, education

It can be seen in the SPSS 'Model Summary' table that the multiple R was estimated at .361, which corresponds to a multiple R^2 value of .130. Thus, 13.0% of the variance in annual earnings was accounted for by a regression equation which included both years of education completed and IQ as predictors. There is also an adjusted R^2 value reported which corresponded to .123. The adjusted R^2 represents a more accurate estimate of the effect in the population (you can square root the adjusted R^2 to get the adjusted model R). Thus, from this more accurate perspective, 12.3% of the variance in earnings would be expected to be accounted for in the population by the multiple regression equation. Finally, the standard error of the estimate was estimated at \$40,375.37, which corresponds to the standard deviation associated with the residuals (i.e., difference between observed dependent variable scores and predicted dependent variable scores). A standard deviation in residuals equal to \$40,375 is massive, considering the mean in annual earnings was only \$43,314. Thus, on average, the prediction equation makes an error in predicting Y (the dependent variable, annual earnings) equal to \$40,375. This is the reality of doing science, in most cases. Recall that the percentage of variance accounted in earnings was only about 13%, so a massive 87% of the variance was not accounted for by education and IQ. Still, it is a numerical improvement over the inclusion of only education as a predictor, which yielded an adjusted model $R^2 = .090$ (reported in chapter 12).

Next up is the 'ANOVA' table which tests the statistical significance of the multiple R value:

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	6.040E+10	2	3.020E+10	18.525	.000 ^b
	Residual	4.027E+11	247	1630170315		
	Total	4.631E+11	249			

a. Dependent Variable: earnings

b. Predictors: (Constant), IQ, education

It can be observed in the 'ANOVA' table that the multiple R value of .361 was associated with an F -value of 18.525. With 2 and 247 degrees of freedom, the F -value was statistically significant, $p < .001$. Note that the F -value corresponds to the unadjusted R value (not the adjusted R value; SPSS does not report an F -value for the adjusted R^2 estimate).

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Next, SPSS reports a table entitled ‘Coefficients’ which includes the values associated with the multiple regression equation.

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	95.0% Confidence Interval for B	
		B	Std. Error	Beta			Lower Bound	Upper Bound
1	(Constant)	-56261.421	17654.882		-3.19	.002	-91035	-21488
	education	3756.289	1169.950	.232	3.211	.002	1451.94	6060.64
	IQ	501.978	208.972	.174	2.402	.017	90.383	913.573

a. Dependent Variable: earnings

It can be observed in the table entitled ‘Coefficients’ that the intercept (or what SPSS refers to as the ‘Constant’) was equal to -56261.42 which is an impossible amount of money to earn in a year (no matter how badly someone does their job). Also, note that none of the values in the data file were negative. It is simply a reality that multiple regression will sometimes estimate an “impossible” intercept value, if that is the best way to come up a multiple regression equation that maximally predicts the dependent variable based on the data.

In the same column as the intercept are the unstandardized beta weights. It can be observed that education was associated with an unstandardized beta weight of $b = 3756.289$. Therefore, a one unit increase in education (i.e., one extra year of education completed) was associated with an approximate unique (not shared with IQ) increase of \$3,756 in annual earnings. To determine if the unstandardized beta weight was statistically significant, you need to look at the ‘Sig.’ column. You’ll see that the education beta weight was associated with a p -value of .002, based on a t -value of 3.211. Furthermore, the unstandardized beta weight of 3756.289 was associated with 95% confidence intervals equal to 1451.938 and 6060.639.

The IQ independent variable was also a statistically significant contributor to the regression equation: $t = 2.402$, $p = .017$. Furthermore, the unstandardized beta weight was equal to $b = 501.978$. Thus, a one unit increase in IQ was associated with an approximate unique (not shared with education) increase of \$501.98 in annual earnings. The 95% confidence intervals for the IQ unstandardized beta weight were equal to 90.383 and 913.573. I haven’t mentioned anything about the ‘Std. Error column’. The main interesting bit about that column is that if you divide the unstandardized beta weight by its corresponding standard error, you get the t -value. So, for IQ, $501.978 / 208.972 = 2.402$.

In this example, reporting and interpreting the unstandardized beta weights made a lot of sense, because they were naturally interpretable. People know what an extra year of education implies, for example. Also, people understand annual earnings expressed as dollars. However, all too often, the unstandardized beta weights are not naturally interpretable, because the independent variables were measured on an unnatural metric (e.g., sum of items

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scored on a 5-point Likert scale), therefore, the standardized beta weights become very useful to interpret in such cases. Also, although the education and IQ unstandardized beta-weights are readily interpretable, they can't be compared in terms of size (i.e., relative contribution to the regression equation). One must consult the standardized beta-weights for such purposes.

In this example, education was associated with a standardized beta weight of $\beta = .232$. You cannot square a standardized beta weight and speak of percentage of variance accounted for. However, you can say that a one standard deviation increase in years of education completed ($SD = 2.67$) corresponded to .232 of a standard deviation increase in annual earnings (i.e., $43123.58 * .232 = \$10,004.67$). Note that, in the context of this regression equation, the .232 of a standard deviation increase in annual earnings is unique to years of education, i.e., not shared with IQ (same fact applies to the unstandardized beta weight). With respect to the IQ standardized beta weight results, a one standard deviation increase in IQ ($SD = 14.93$) was associated with a .174 increase in annual earnings (i.e., $43123.58 * .174 = \$7503.50$). In practice, the statistical significance associated with standardized beta weights is implied to be consistent with the unstandardized beta weights. Therefore, researchers simply interpret a standardized beta weight as statistically significant, if the unstandardized beta weight was found to be statistically significant (**Watch Video 14.2: [How to interpret a standardized beta weight](#)**).

As mentioned, one of the benefits associated with standardized beta weights is that you can compare their importance in a regression equation. For example, the standardized beta weight associated with IQ was equal to .174, which was *numerically* smaller than the standardized beta weight of .232 associated with education. Thus, at least numerically, years of education completed was a more substantial, unique contributor to the regression equation than IQ. Although researchers very frequently interpret a numerical difference between two standardized beta weights as statistical support for interpreting one predictor as a more importance contributor to the model, the reality is that such numerical differences can easily arise due to sampling fluctuations. Consequently, it is necessary to test the numerical difference between two standardized beta weights statistically, before making any interpretations. Unfortunately, very few, if any, programs test the difference between two standardized beta weights statistically. In the Advanced Topics section of this chapter, I demonstrate an arguably appropriate technique to test the difference between two standardized beta weights in SPSS, using bootstrapping and confidence interval overlap estimation.

Common Mistakes in Interpreting Standardized Beta Weights

Although a standardized beta weight does share some similarities to a Pearson correlation, it falls short in one important respect: a standardized beta weight in multiple regression cannot be converted into a coefficient of determination. That means that you can't

square the beta weight and speak of the percentage of variance in the dependent variable accounted for by the independent variable. In this context, a researcher has the option of consulting and reporting squared semi-partial (type A) correlations, instead. Consequently, I recommend that you always report both the standardized beta weights and the semi-partial (type A) correlations in a multiple regression analysis.³

Another common mistake is to state that the largest standardized beta weight is the most significant predictor of the dependent variable. In order to make such a statement justifiably, one would need to test the difference between two beta weights statistically. It is not enough to simply look at the numerical difference between two weights and assert one to be the most significant predictor of a dependent variable.

Finally, all too often, researchers fail to specify whether they have reported the unstandardized (b) or standardized (β) beta weights in their results. In the behavioral sciences, it usually only makes sense to report the standardized beta weights, as the variables are typically measured on an inherently meaningless scale. In some disciplines, however, many of the independent and dependent variables are associated with meaningful scales (e.g., economics). Consequently, the unstandardized beta weights can be interpreted meaningfully in such cases. Ultimately, it is up to the researcher to decide whether it is best to report the unstandardized or standardized beta weights (or both). So long as the beta weights are demarcated clearly as either unstandardized (b) or standardized (β), the reader should be able to follow the results coherently.

Multiple Regression and Semi-Partial Correlations

Although beta weights are informative, they lack the capacity to be squared. Consequently, researchers cannot speak of percentage of variance in the dependent variable accounted for by the independent variables. To overcome this limitation in beta weights, researchers should report the semi-partial correlations (Type A) associated with the independent variables included in the multiple regression model. The multiple regression utility in SPSS allows for the estimation of both partial and semi-partial correlations (**Watch Video 14.3: [Semi-Partial Correlations & Multiple Regression in SPSS](#)**). The key portion of the output is included in the SPSS table entitled 'Coefficients':

³Journal articles commonly include only the standardized beta weights. Fortunately, beta-weights can be converted by a reader into squared semi-partial correlations, based on a simple re-arrangement of the formula specified by Tzelgov and Stern (1978, p. 330). Thus, to derive a semi-partial correlation from a beta-weight, the following formula can be used: $r_{Y1.2} = \beta_{Y1.2} * (\sqrt{1 - r_{12}^2})$

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Coefficients ^a									
		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Correlations		
		B	Std. Error	Beta			Zero-order	Partial	Part
1	(Constant)	-56261....	17654.88		-3.187	.002			
	education	3756.289	1169.950	.232	3.211	.002	.332	.200	.190
	IQ	501.978	208.972	.174	2.402	.017	.307	.151	.143

a. Dependent Variable: earnings

SPSS refers to semi-partial correlations (i.e., Type A) as Part correlations. As you can see in the SPSS table entitled 'Coefficients', education was associated with a type A semi-partial correlation of $r = .190$, or $r^2 = .036$. Thus, 3.6% of the variance in annual earnings was accounted for uniquely by years of education completed. IQ was associated with a type A semi-partial correlation of $r = .143$, or $r^2 = .020$. Thus, 2.0% of the variance in annual earnings was accounted for uniquely by IQ. It only really makes sense to report type A semi-partial correlations in the context of multiple regression, not the type B semi-partial correlations (see Chapter 13). The statistical significance associated with the type A semi-partial correlations is implied to be the same as the unstandardized beta weights. Consequently, there is no additional 'Sig.' column for the additional correlations included in the 'Coefficients' table.

I think it is good practice to always report the semi-partial correlations associated with the independent variables included in a multiple regression model. An additional interesting attribute associated with the squared semi-partial correlations is that they can be summed. In this example, the sum of the squared semi-partial correlations amounted to: $.036 + .020 = .056$. Thus, 5.6% of the variance in earnings was accounted for by education and IQ, uniquely. However, you may recall that the multiple R^2 was equal to .130, or 13.0%. What can account for the difference between 13.0% and 5.6%? Well, the difference of 7.4% ($13.0 - 5.6$) may be described as the percentage in earnings variance that was accounted for by education and IQ in a shared manner, i.e., not uniquely. Recall that, in this example, education and IQ shared 32.7% ($r^2 = .327$) of their variance. That shared education and IQ variance evidenced an impact on the model R^2 equal to 7.4%.

Multiple Regression Assumptions

Multiple regression is a parametric statistical analysis, consequently, it is associated with a number of assumptions. All of the assumptions are the same as those described in the chapter devoted to bivariate regression, in addition to one assumption unique to multiple regression: the absence of multi-collinearity. Consequently, the assumptions in common with bivariate regression will only be listed and tested here, rather than described in a substantial amount of detail.

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1: Random sampling

The assumption implies that every case in the population has an equal chance of being selected into the sample. The data were not obtained via random sampling in Zagorsky (2007). The impact of such a violation on the results is anyone's guess.

2: Independence of Observations

See bivariate regression chapter.

3: Dependent Variable Measured on an Interval/Ratio Scale

The assumption that the dependent variable is measured on an interval/ratio scale has been satisfied, as annual earnings (DV) was measured on a ratio scale.

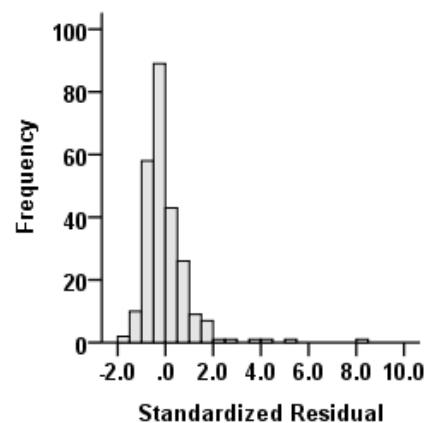
4: Linearity

See bivariate regression chapter.

5: Normally distributed residuals

As can be seen in Figure C14.1, the assumption that the residuals would be associated with a normal distribution was not satisfied. Specifically, the skew and kurtosis associated with the residuals above was 3.49 and 21.00, respectively, which is very substantial. To my knowledge, in the context of multiple regression, there is no precise rule about how skewed or kurtotic a distribution of residuals can be, before it becomes problematic. Perhaps the rule of skew < 2.0 and kurtosis < 9.0 might be applicable here. Fortunately, a multiple regression analysis can be performed with bootstrapping as an estimation procedure, rather than asymptotic normal distribution theory. Bootstrapping does not assume any level of normality, so, it can be potentially applicable to any set of data. I demonstrate a bootstrapped multiple regression in the Advanced Topics section of this chapter.

Figure C14.1. Histogram of Standardized Residuals



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6: Independence of errors

See the bivariate regression chapter.

7: Homoscedasticity

See the bivariate regression chapter for a full discussion of this assumption. I tested the homoscedasticity assumption by estimating the correlation between the standardized predicted values and the absolute standardized residuals. The Pearson and Spearman correlations were $r = .17$ ($p = .006$) and $r_s = .14$ ($p = .028$), respectively. The statistically significant correlations suggest that the homoscedasticity assumption has not been satisfied (**Watch Video 14.4: [Testing Homoscedasticity via Correlation](#)**). In the Advanced Topics section of this chapter, I describe a more commonly used procedure to test the assumption of homoscedasticity known as the Koenker test. Unfortunately, the Koenker test cannot be applied within the menu options of SPSS. I also describe in the Advanced Topics section of this chapter a procedure that can be implemented to deal with the presence of heteroscedasticity.

8: Absence of Multi-collinearity

Strictly speaking, the absence of multicollinearity is not really an assumption of multiple regression, as the results obtained from a multiple regression analysis in the presence of multicollinearity will be accurate. However, it is still important to pay attention to the possibility that multicollinearity may be present in one's data, as the stability of the model will be affected negatively.

In the context of multiple regression, I like to distinguish between collinearity and multicollinearity. Collinearity is observed when one independent variable correlates approximately $r = .95$ or higher with another independent variable. By contrast, multicollinearity is observed when one independent variable is associated with a multiple correlation of $R = .95$ or higher, based on a prediction equation with the other independent variables included in the multiple regression analysis. Thus, collinearity can be evaluated with a simple examination of the Pearson correlation matrix. By contrast, multicollinearity cannot be evaluated with a simple eye-ball examination of the correlation matrix. Instead, more sophisticated analyses need to be performed such as Tolerance and/or the Variance Inflation Factor. I describe these next.

Tolerance

Tolerance represents the percentage of variance in a particular independent variable that is not shared with the other independent variables included in the multiple regression analysis. Stated alternatively, tolerance represents the percentage of variance that is unique to a particular independent variable. In multiple regression, it is desirable for all of the independent variables to be associated with some unique variance. The percentage of

variance not shared with the other independent variables is estimated by regressing the independent variable onto the remaining independent variables. Consequently, tolerance may be formulated as:

$$\text{Tolerance} = 1 - R^2$$

Tolerance values are calculated for each independent variable included in the analysis. Therefore, if five independent variables are included in the model, then five tolerance values will be estimated.

All other things equal, researchers desire higher levels of tolerance. Tolerance values of .10 or greater are commonly considered acceptable. Such a value would imply that each independent variable requires a minimum of 10% unique variance. It would also imply that no independent variable is associated with a multiple R of .95 or greater (i.e., $.95^2 = .90$).

Variance Inflation Factor

The variance inflation factor represents the degree to which a beta weight's standard error is larger than it would otherwise be, under the pretense that the particular independent variable is uncorrelated completely with the other independent variables included in the analysis. Not many people seem to be aware of this, but the variance inflation factor (VIF) is correlated perfectly with tolerance. Specifically, the VIF is the reciprocal of tolerance:

$$\text{VIF} = \frac{1}{\text{Tolerance}}$$

Consequently, a recommendation for an acceptable value of tolerance needs to be consistent with a recommendation for the VIF. A minimum tolerance value of .10 was recommended above, which would imply a maximum VIF of 10. Some have recommended a maximum VIF as small as 4 (Pan & Jackson, 2008), which would imply a minimum tolerance of .25. Like so many things in statistical life, there is no real or truly meaningful cut-off value. Personally, I'm fine with a VIF as large as 10, if the sample size is relatively large (> 100). Sample size comes into play here, as larger standard errors associated with beta weights reduces the chances of observing the corresponding beta weights as statistically significant.

SPSS allows for the estimation of both Tolerance and VIF through the linear regression utility (**Watch Video 14.5: [Evaluating Multicollinearity \(Tolerance & Variance Inflation Factor\) in SPSS](#)**). SPSS adds the Tolerance and VIF statistics at the right-side end of the 'Coefficients' table:

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Coefficients^a

Model	Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics	
	B	Std. Error	Beta			Tolerance	VIF
1 (Constant)	-56261.421	17654.882		-3.187	.002		
education	3756.289	1169.950	.232	3.211	.002	.672	1.487
IQ	501.978	208.972	.174	2.402	.017	.672	1.487

a. Dependent Variable: earnings

It can be observed that education and IQ were both associated with the same Tolerance and VIF values, as expected, because there were only two independent variables in the analysis. When there are three or more independent variables included in the analysis, the tolerance/VIF values will almost always be different across the independent variables. In this example, education and IQ were associated with tolerance of .672, which implies that they were associated with 67.2% unique variance, i.e., not predicted by the other independent variable. Such a value exceeds comfortably the minimum acceptable level of tolerance of .10. Correspondingly, the VIF was equal to 1.487 ($1/.672 = 1.488$), which implies that the beta weight standard errors were approximately 1.488 times larger than they would otherwise be, if there was no correlation between the two independent variables included in the analysis. As the VIF value of 1.488 is much less than the maximum acceptable VIF value of 10, the issue of multicollinearity (or collinearity, more precisely, in this case) is not a problem, in this example.

Dealing with Multi-Collinearity

The main point of understanding multi-collinearity is that very high levels present in your data will thwart the chances of identifying a beta-weight as statistically significant. In practice, there will probably be not much lost by dumping an independent variable from the analysis, if the independent variable is associated with a large VIF (10+), as the other predictors will pick-up the variance in the dependent variable that was predicted by that particular independent variable. However, it should be noted that one does not desire or expect extremely low levels of VIF either, because one of the main reasons one conducts a multiple regression analysis is to understand the nature of the effects of two or more inter-related independent variables on a dependent variable. If the independent variables were not inter-related ($VIF = 0$), there would not be much point in conducting the multiple regression analysis in the first place. For more sophisticated approaches to the evaluation of multi-collinearity, one might consider condition indices in conjunction with variance-decomposition proportions (Belsey, Kuh, & Welsch, 2004; see also O'Brien, 2007). SPSS does output such information after the coefficients table, when the 'Collinearity diagnostics' option is selected. However, I believe that consulting the VIF/tolerance values should suffice, in most cases.

Sample Size Requirements

Researchers commonly ask the question of how many cases they need in order to conduct a multiple regression. Over the years, a number of rules of thumb have emerged. For example, Tabachnik and Fidell (1989) recommended a bare minimum of five cases for each predictor included in the analysis. Therefore, a multiple regression with four predictors would require a bare minimum of 20 cases. Wampold and Freund (1987) recommended a minimum of 10 cases per predictor. Finally, Nunnally (1978) advised that a minimum sample size of 300 to 400 would be required, in order to obtain accurate results for a typical multiple regression.

First and foremost, it needs to be acknowledged that the issue of sample size and multiple regression is a question of statistical power. Consequently, no rule of thumb will ever be appropriate in all cases, or even many cases. Secondly, one must appreciate that sample size requirements will depend, in part, on the purpose of conducting the multiple regression. Specifically, is it simply to estimate a multiple R^2 value? Or is it to evaluate statistically two or more hypothesized predictors of a dependent variable? As you will see below, if the purpose is to address the second question, then much greater sample sizes are required.

Green (1991) evaluated the sample size requirements for multiple regression by explicitly framing the problem within the context of power. Based on his work, Green (1991) specified that, if the purpose of conducting the multiple regression is to simply estimate a multiple R^2 value, then the following guideline could be used:

$$N = 50 + 8 * p \quad (2)$$

where p equals the number of predictors included in the analysis. So, for example, if one wished to conduct a multiple correlation analysis with four predictors, a sample size of 82 would be required, based on Green's (1991) guideline. Bear in mind that Green's (1991) guideline assumes that a medium sized multiple R^2 value of .07 exists in the population. If a smaller multiple R^2 value is hypothesized to exist in the population, then a larger sample size would be required. Correspondingly, a larger expected multiple R^2 value in the population would entail a smaller required N for the same level of power (i.e., .80).⁴

In practice, very few researchers are simply interested in the multiple R^2 value associated with a multiple regression analysis. Instead, the vast majority of researchers are interested in evaluating the statistical significance associated with their hypothesized predictors included in the model, in addition to the multiple R^2 . In such cases, again, based on a hypothesized medium effect size present in the population, Green (1991) proposed the following guideline:

$$N = 105 + p$$

⁴ Power represents the probability of rejecting the null hypothesis when it is, in fact, false in the population. A minimum power level of .80 (or 80%) has been recommended (Cohen, 1988).

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Thus, based on Green (1991), realistically, a typical multiple regression analysis would require a minimum of approximately 105 participants. So, for example, if one wished to conduct a multiple correlation analysis with four predictors, and one was interested in examining the beta weights, a sample size of 109 would be required, based on Green's (1991) guidelines. Green's (1991) guidelines appear to have become fairly popular over the years, based on the number of times his paper has been cited.

More recently, Maxwell (2000) extended Green's (1991) research by pointing out that the magnitude of the correlations between the predictors will affect the sample size required to conduct a multiple regression with a specified level of power. Maxwell's (2000) work is directly related to the variance inflation factor described in the context of multi-collinearity. You'll recall that, as the correlations between the predictors in a multiple regression increase, so do the standard errors associated with the predictor beta weights. Consequently, Maxwell (2000) reasoned, larger correlations between the predictors implies the need for greater samples sizes, all other things equal, if one is interested in both the beta weights and the multiple R . Maxwell (2000) conducted an extensive simulation study to estimate the sample sizes required to detect statistically significant beta weights across a variety of conditions, including the average magnitude of the correlations between the predictors, and a specified power of .80. I have provided the table below which includes some of the key results reported by Maxwell (2000) in Table C14.1.

Table C14.1. Sample Sizes Required for Multiple Regression (5 predictors; Power = .80)

		p_{xy}	
p_{xx}		.30	.40
.30	1,070	419	191
.40	1,731	692	328
.50	2,752	1,117	544

Note. Based on results reported in Maxwell (2000); p_{xx} = mean inter-correlation between predictors; p_{xy} = mean inter-correlation between predictors and dependent variable.

Clearly, very large sample sizes are required, when the correlations between the predictors are large. For example, if the average correlation between the predictors and the dependent variable is $r = .30$ and the average correlation between the predictors is $r = .30$, a sample size of 1,117 is required to achieve power of .80. However, if the number of predictors is reduced to three, the sample size required to achieve power of .80 for the entire analysis is reduced to a much more manageable 218 (see Table 5 in Maxwell, 2000). I used the words 'entire analysis', because Maxwell (2000) estimated power under the pretense that the researcher was interested in testing the model R for statistical significance, as well as each of the predictors (i.e., not at least one predictor).

C14.17

Hierarchical Multiple Regression

The simplest approach to conducting a multiple regression is known as “method enter”, as it forces all of the independent variables into the model, whether they are statistically significant contributors to the regression equation or not. The preceding demonstration was performed in the context of method enter multiple regression.

A slightly more sophisticated approach to multiple regression analysis is known as ‘hierarchical multiple regression’. Hierarchical multiple regression is performed in a sequence specified by the researcher. Essentially, the independent variables are specified to be included into the model in a series of blocks. Each block of independent variable(s) is added to the model, and the increase in model R^2 as a result of adding the block is tested for statistical significance. Each block may have one or more independent variables.

In practice, researchers typically use hierarchical multiple regression such that they add one or more *less* interesting independent variables in the first block, and then add one or more theoretically interesting independent variables in subsequent blocks. My impression is that, typically, there are only two blocks in published hierarchical multiple regression analyses. The first block will include “control” variables, such as socioeconomic status, age, and gender, for example. Then, the theoretically more interesting variables are added in the subsequent block(s).

In the context of the years of education, intelligence, and earnings example, it may be suggested that education is a relatively well established predictor of earnings. By contrast, the hypothesis that intelligence may contribute to a regression equation in a unique manner is less well-established and, arguably, more interesting, theoretically. Thus, a researcher may conduct a hierarchical multiple regression analysis with years of education completed entered in block 1 and intelligence entered in block 2. The hypothesis that intelligence can add to the regression model in a statistically significant manner would be tested by the R^2 change observed from block 1 to block 2. If the R^2 squared changes statistically significantly from block 1 to block 2, then intelligence would be determined to be a statistically significant, unique contributor to the model.

Hierarchical Multiple Regression: SPSS

In order to conduct a hierarchical multiple regression in SPSS, one can use the regular linear regression utility (**Watch Video 14.6: [Hierarchical Multiple Regression in SPSS](#)**). The first table outputted by SPSS is entitled ‘Variables Entered/Removed’. It can be seen that Model 1 includes ‘education’ as an entered variable. Although not obvious, Model 2 includes both education and IQ.

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Variables Entered/Removed^a

Model	Variables Entered	Variables Removed	Method
1	education ^b	.	Enter
2	IQ ^b	.	Enter

a. Dependent Variable: earnings

b. All requested variables entered.

Next, in a SPSS table entitled 'Model Summary', you will find the model R , model R^2 , model adjusted R^2 values for both Model 1 and Model 2. It can be seen that with the addition of education to the model (Model 1), the model R^2 was equal to .110. By contrast, with the addition of IQ to the model (education and IQ together), the model R^2 increased to .130. On the right side of the 'Model Summary' table you will find the 'change statistics'. These statistics correspond to the changes in model R^2 values and their corresponding statistical significance. The change statistics for model 1 correspond to no independent variables included in the model to the addition of one independent variable, which is education in this example. The R^2 change from .000 to .110 was statistically significant, $F(1, 248) = 30.69$, $p < .001$. Thus, the percentage of variance accounted for earnings by the inclusion of education into the model was statistically significant.

The R^2 change associated with model 2 (i.e., the addition to intelligence to the model) corresponded to .020 and it was statistically significant, $F(1, 247) = 5.77$, $p = .017$. Thus, the addition of intelligence to the model increased the percentage of variance accounted for in earnings in a statistically significant and unique manner. Clearly, it was not a large increase, but it was a statistically significant one.

Model Summary^c

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics				
					R Square Change	F Change	df1	df2	Sig. F Change
1	.332 ^a	.110	.107	40761.8223	.110	30.690	1	248	.000
2	.361 ^b	.130	.123	40375.3677	.020	5.770	1	247	.017

a. Predictors: (Constant), education

b. Predictors: (Constant), education, IQ

c. Dependent Variable: earnings

Next, a table entitled 'ANOVA' includes the F and p -values associated with the model R^2 values reported in the 'Model Summary' table. Model 1's R^2 value of .110 was statistically significant, $F(1, 248) = 30.69$, $p < .001$. Additionally, model 2's R^2 value of .130 was statistically significant, $F(2, 247) = 18.53$, $p < .001$.

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ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	5.099E+10	1	5.099E+10	30.690	.000 ^b
	Residual	4.121E+11	248	1661526160		
	Total	4.631E+11	249			
2	Regression	6.040E+10	2	3.020E+10	18.525	.000 ^c
	Residual	4.027E+11	247	1630170315		
	Total	4.631E+11	249			

a. Dependent Variable: earnings

b. Predictors: (Constant), education

c. Predictors: (Constant), education, IQ

Next, a table entitled 'Coefficients' includes all of the statistics relevant to the individual predictors.

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Correlations		
		B	Std. Error	Beta			Zero-order	Partial	Part
1	(Constant)	-27226.696	12991.618		-2.096	.037			
	education	5365.130	968.456	.332	5.540	.000	.332	.332	.332
2	(Constant)	-56261.421	17654.882		-3.187	.002			
	education	3756.289	1169.950	.232	3.211	.002	.332	.200	.190
	IQ	501.978	208.972	.174	2.402	.017	.307	.151	.143

a. Dependent Variable: earnings

Hierarchical Multiple Regression: Comments

You may have noticed that the R^2 change p -value associated with the addition of block 2 to the model (i.e., intelligence predictor) was exactly the same as the p -value associated with the intelligence beta weight p -value. Consequently, the R^2 change statistical analysis result and the beta weight analysis result were redundant with each other. This will always be the case when one variable is included in a block. Ultimately, with just two predictors included in the whole model, a hierarchical multiple regression will not really yield any more informative results than a method enter multiple regression. I used this example, here, for the sake of simplicity and continuity. A more cogent example might include education, socioeconomic status, age, and gender as predictor variables entered in block 1, and then IQ and perseverance entered in block 2. The key is to have two or more predictor variables in each block. In such cases, the R^2 change p -value will unlikely correspond to any of the predictor p -values associated with a block. Thus, the test of R^2 change can yield important non-redundant information, in such a case.

Stepwise Multiple Regression

In some cases, you have a research scenario where you have a large number of independent variables to include in a multiple regression analysis. Furthermore, the purpose of the analysis may be to identify a smaller number of variables that predict the dependent variable in a unique way, i.e., each independent variable contributes at least some statistically significant variance to the regression equation. In such cases, it has been argued that stepwise multiple regression can be a useful strategy to conducting the multiple regression analysis (Pope & Webster, 1972).

The steps involved in conducting a stepwise multiple regression are purely statistical and “automated”. That is, the process of selecting which predictor variables that will be included in the regression equation is entirely algorithmic. At step 1, the stepwise analysis will identify the independent variable that has the largest zero-order Pearson correlation with the dependent variable. That variable will be added to the regression model as step 1. Then, the stepwise procedure will look for the independent variable that has the largest partial correlation with the dependent variable, controlling for the variance accounted for by the first independent variable selected at step 1. If the second independent variable’s first-order partial correlation is found to be significant statistically, the second independent variable will be added to the regression equation. Next, the analysis looks for the independent variable with the largest second-order partial correlation amongst the remaining independent variables, controlling for the variance accounted for by the first two predictors (i.e., included in the regression equation) and the candidate third predictor. If the second-order partial correlation is statistically significant, the stepwise procedure will add the predictor to the model. Eventually, the procedure is terminated, once a non-significant partial correlation is identified as the next largest.

It is important to note that, at each step, previously entered predictors are evaluated to determine whether they have maintained their statistically significant contribution to the regression equation. If a previously entered variable loses its statistically significant contribution, it will be dropped from model. Consequently, you may come across an example where the first variable entered into the equation/model, i.e., the one with the largest zero-order correlation with the dependent variable, is not included in the final regression equation. This would happen if that first variable shared a fairly substantial amount of variance with the other independent variables. So, just because an independent variable is not included in a stepwise multiple regression equation, you should not claim that it is unrelated to the dependent variable: it may, in fact, have a large, or even the largest, Pearson correlation with the dependent variable.

In practice, a stepwise multiple regression may involve adding 10 candidate independent variables to the multiple regression analysis; however, only 2 or 3 are kept in the

regression equation, because they are the only ones that are found to be significant statistically and unique contributors to the regression equation.

Stepwise Multiple Regression Example: SPSS

Heart disease is one of the most substantial culprits of death amongst adult humans. Much research money continues to be poured into discovering why heart disease develops, as well as variables that predict heart disease. One commonly discussed predictor of heart disease is cholesterol. Although not everyone agrees⁵, individual differences in cholesterol levels is commonly considered a predictor of heart disease. Consequently, many people are recommended by their medical practitioners to obtain a cholesterol test, which involves getting a blood test and some analysis of the blood. Another doctor's appointment is typically required to obtain the results. This is expensive when multiplied across millions and millions of people around the world. Imagine if cholesterol levels could be predicted with appreciable accuracy, based on variables that are easily and inexpensively assessed.

Based on the results reported by Bonaa and Thelle (1991), in addition to a couple of other studies, I have simulated ($N = 1,000$) some data (**Date File: predicting_cholesterol**) to correspond to a possible study designed to try to predict cholesterol levels with the following candidate independent variables: age, sex, blood pressure (SBP), body mass index (BMI), cigarette smoking (Cig), physical activity (PhysAct), alcohol consumption (Alc), family history of cholesterol problems (famhistory), and hours of sleep per day (sleep). Thus, a total of nine candidate independent variables. One approach to predicting cholesterol would be to simply enter (force) all nine independent variables into the model. However, such an approach would almost certainly lead to the inclusion of several non-significant contributors to the equation. This is where stepwise multiple regression could be potentially used. That is, why not let the statistical algorithm decide which variables to enter and exclude in the model on the basis of the correlations between the variables?

When I conducted the analysis in SPSS, I obtained the key results below (**Watch Video 14.7: [Stepwise Multiple Regression in SPSS](#)**). As can be seen in the SPSS table entitled 'Model Summary', seven models were estimated; furthermore, each successive model incorporated an additional predictor. In the first model, 'age' was included as the predictor variable, because it had the largest zero-order correlation with the dependent variable ($r = .44$). The model R^2 was .194, thus, 19.4% of the variance in cholesterol was accounted for by age alone. Next, the stepwise algorithm selected 'sleep' as the next variable to include in the model, because it had the largest partial correlation with the dependent variable. It can be seen that the model R^2 increased by .070, which implies that an additional 7% of the variance in

⁵ You might be surprised to learn just how much disagreement there is among respected researchers with respect to the role of cholesterol in heart disease.

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cholesterol was accounted for by the inclusion of sleep. The stepwise analysis terminated after the inclusion of seven predictor variables, which accounted for 36.0% (model $R^2 = .360$) of the variance in cholesterol. Also note that the final variable included in the model ('sex') increased the percentage of variance accounted for by only 1% (R^2 change = .010).

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics				
					R Square Change	F Change	df1	df2	Sig. F Change
1	.440 ^a	.194	.193	.89845	.194	239.599	1	998	.000
2	.513 ^b	.264	.262	.85896	.070	94.880	1	997	.000
3	.551 ^c	.304	.302	.83544	.040	57.909	1	996	.000
4	.573 ^d	.328	.325	.82135	.024	35.461	1	995	.000
5	.584 ^e	.341	.337	.81403	.013	18.989	1	994	.000
6	.591 ^f	.350	.346	.80890	.009	13.653	1	993	.000
7	.600 ^g	.360	.355	.80295	.010	15.750	1	992	.000

a. Predictors: (Constant), Age

b. Predictors: (Constant), Age, sleep

c. Predictors: (Constant), Age, sleep, famhistory

d. Predictors: (Constant), Age, sleep, famhistory, BMI

e. Predictors: (Constant), Age, sleep, famhistory, BMI, Cig

f. Predictors: (Constant), Age, sleep, famhistory, BMI, Cig, SBP

g. Predictors: (Constant), Age, sleep, famhistory, BMI, Cig, SBP, sex

Like any other hierarchical-type regression, SPSS then reports a table entitled 'ANOVA', which tests for statistical significance each of the Model R values (from Model 1 $R = .440$ to Model 7 $R = .600$). I have not reported the SPSS table of results, here, as the table is large. Suffice it to say that all of the Model R values were found to be significant statistically. Next, SPSS reports the beta-weights associated with the variables included in the model. As can be seen in the SPSS table entitled 'Coefficients', age was associated with a standardized beta-weight of .430 and the largest t -value. By comparison, sex (coded 1 = female; 2 = male) was associated with a standardized beta-weight of .102. Most of the semi-partial correlations are of a similar magnitude to the standardized beta-weights, as the independent variables were not particularly inter-correlated. An exception to that statement was BMI.

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Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Correlations		
		B	Std. Error	Beta			Zero-order	Partial	Part
7	(Constant)	-.861	.114		-7.578	.000			
	Age	.430	.027	.430	16.218	.000	.440	.458	.412
	sleep	.270	.026	.270	10.298	.000	.210	.311	.262
	famhistory	.369	.051	.185	7.223	.000	.194	.224	.183
	BMI	.141	.027	.141	5.231	.000	.290	.164	.133
	Cig	.111	.026	.111	4.331	.000	.090	.136	.110
	SBP	.108	.027	.108	4.054	.000	.140	.128	.103
	sex	.204	.051	.102	3.969	.000	.114	.125	.101

a. Dependent Variable: TC

Next, SPSS reports the independent variables excluded from the regression equation in a table entitled, appropriately enough, 'Excluded variables'. Again, because the table is so long, I reported only the top and bottom of the table. It can be seen that, in the final model (model 7), two variables were left out of the regression equation: PhysAct (Physical Activity) and Alc (Alcohol consumption). It can be seen that the partial correlations are rather small at $-.062$ and $.016$, respectively, neither of which were significant statistically. Consequently, they were omitted from the regression equation.

Excluded Variables^a

Model		Beta In	t	Sig.	Partial Correlation	Collinearity Statistics
						Tolerance
4	Age	.442 ^h	1.822	.033	.422	.922
7	PhysAct	-.050 ^h	-1.951	.051	-.062	.985
	Alc	.014 ^h	.517	.605	.016	.886

Thus, with respect to the prospect of predicting a person's cholesterol level without taking a blood test, more work needs to be done, as only 36% of the variance in cholesterol was accounted for by the regression equation that included seven predictors. Naturally, many other variables could be potentially added to the regression equation, which may increase the model R^2 . Conceivably, hundreds of variables could be added to the regression equation to help increase model R^2 . With a sufficient sample size (in the hundreds of thousands), the results may prove robust. Who knows? Maybe one day.

Importantly, it would be unjustified to suggest that the variables omitted from the regression equation were unrelated to cholesterol levels. As can be seen in the Pearson correlation table below, both PhysAct and Alc were correlated statistically significantly with cholesterol (all correlations greater than $|.06|$ were significant statistically). The multiple regression results simply suggest that they were not related to the cholesterol in a unique way, i.e., independently of the effects attributable to the other independent variables. I also note that PhysAct was very nearly included in the model with a p -value of $.051$. Of course, another

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sample of 1,000 people may yield a slightly different correlation matrix, which may have resulted in PhysAct included in the model and, possibly, sex excluded. Thus, strong arguments about the value of a variable as a predictor should not be made on the basis of a stepwise multiple regression analysis (or any multiple regression analysis for that matter). Large samples sizes and replication is crucial, with respect to interpreting the nature and strength of the predictors. The model R^2 , on the other hand, would be expected to be more stable from sample to sample.

Inter-Item Correlation Matrix

	TC	Age	sex	SBP	BMI	Cig	PhysAct	Alc	famhistory	sleep
TC	1.00	.44	.11	.14	.29	.09	-.11	-.12	.19	.21
Age	.44	1.00	-.01	.12	.25	.00	-.08	-.31	-.03	-.12
sex	.11	-.01	1.00	-.09	.07	.01	.04	-.09	-.03	.08
SBP	.14	.12	-.09	1.00	.20	.00	-.01	.01	.07	-.19
BMI	.29	.25	.07	.20	1.00	-.08	-.06	-.07	.06	.04
Cig	.09	.00	.01	.00	-.08	1.0	-.02	-.07	.03	-.06
PhysAct	-.11	-.08	.04	-.01	-.06	-.02	1.00	-.03	-.04	-.04
Alc	-.12	-.31	-.09	.01	-.07	-.07	-.03	1.00	.03	.08
famhistory	.19	-.03	-.03	.07	.06	.03	-.04	.03	1.00	.03
sleep	.21	-.12	.08	-.19	.04	-.06	-.04	.08	.03	1.00

Criticisms of Stepwise Multiple Regression

Few statistical procedures have received more critiques than stepwise multiple regression. Although I can appreciate most of the arguments levelled against the procedure, I believe contentions to the effect that it should never be used are overstated. Virtually all statistical analyses have limitations. Correspondingly, most statistical analyses have a time and place for useful application. From my review, there are a number of arguments that have been articulated against the use of stepwise regression. I discuss a couple of the most important ones here.

First, some have argued that stepwise regression is too empirically driven, with no theory considered in the selection of the variables to the model (e.g., Lewis-Beck, 1978). If researchers had enough theory to specify and test a specific model, I suspect they would do so. Instead, researchers tend to use stepwise multiple regression more as an exploratory approach to the generation of multiple regression equation, because that is the stage of research in which they find themselves. In other cases, a research may be interested exclusively in the solution to an applied problem (e.g., predicting cholesterol levels inexpensively). In my view, that's fine, so long as one is upfront about it. Curiously, exploratory factor analysis (see Chapter 18) relatively rarely gets criticized for lacking theory; so, why criticize exploratory stepwise regression so harshly for lacking theory?

Another criticism of stepwise regression pertains to the inappropriateness of researchers declaring that one predictor is more important than another, with respect to

predicting the dependent variable (Lewis-Beck, 1978). For example, a researcher may state that the first variable entered into the model is more important than the second variable entered into the model. This is a criticism of interpretation, not the stepwise procedure itself. From an inferential statistical perspective, differences between point-estimates (e.g., beta-weights) should be tested statistically. Consequently, it is inappropriate to declare the difference between two beta weights as significant, simply because a stepwise procedure entered them into a model at different steps. Although entirely true, such a criticism can be leveled against *any* approach to multiple regression equation building, from the ‘method: enter’ approach to the fully specified path analytic approach. Consequently, I believe this criticism has also been levelled against the stepwise approach, unfairly.

As a general statement, I would encourage users of stepwise multiple regression to, first, verify that at least two (or more) of the candidate predictors are correlated statistically significantly with the dependent variable, as data sampled randomly from a population of non-correlated variables can yield a regression equation with “statistically significant” ($p < .05$) beta weights (Antonakis & Dietz, 2011). Again, this recommendation could be applied to virtually any multiple regression procedure, unless suppressor effects were hypothesized (in which case, one would probably test suppression directly). My recommendation, here, is consistent with applying Bartlett’s test of sphericity on a correlation matrix, prior to conducting an exploratory factor analysis (Dziuban & Shirkey, 1974).

Curvilinear Multiple Regression

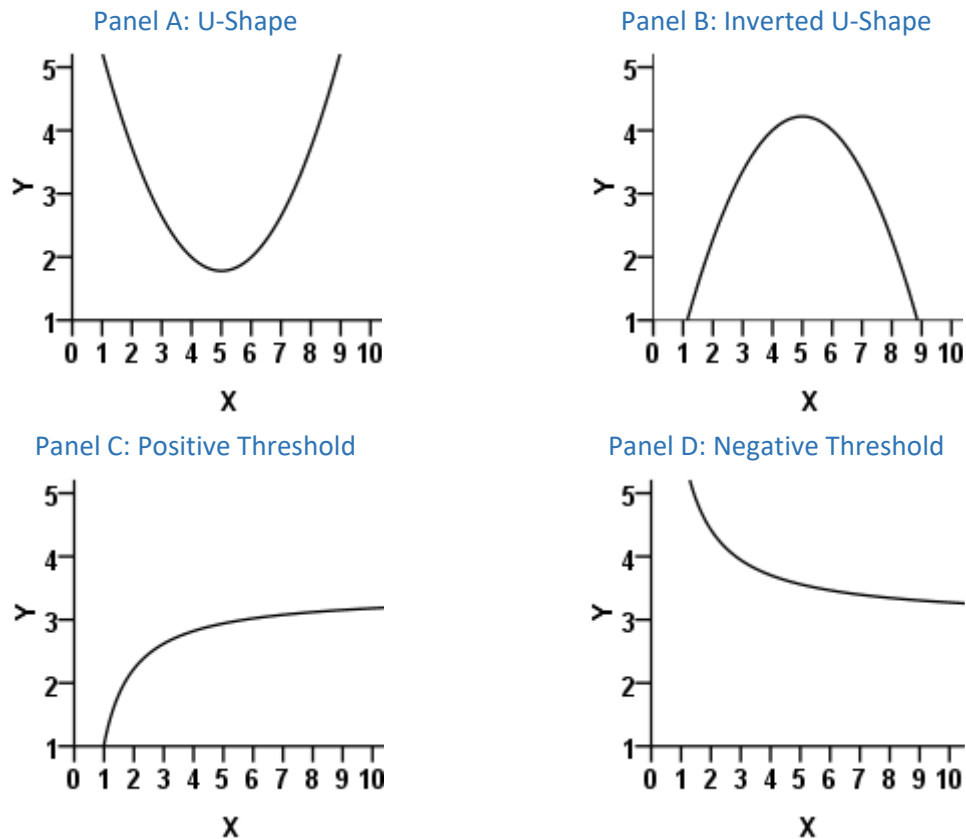
In a previous chapter, it was mentioned that bivariate correlation (e.g., Pearson correlation) and bivariate regression assume that the association between the independent variable and the dependent variable is linear. A linear effect implies that the direction and strength of the association is the same across the whole spectrum of the scores. In some cases, this assumption will not be satisfied. In practice, researchers will examine visually the scatter plot to detect the possibility of the violation of the assumption of linearity. In other cases, however, researchers will generate a hypothesis that the association between the independent and dependent variables is non-linear (a.k.a., curvilinear). The hypothesis may be generated on the basis of theory or previous research. In this section of the chapter, I will demonstrate how to test the most commonly observed curvilinear effect: a quadratic effect.

Pryzbylski and Weinstein (2017) were interested in examining the association between the amount of time adolescents play video games per day and their mental well-being. Based on several considerations, Pryzbylski and Weinstein (2017) were open to the possibility that the association might be curvilinear. That is, they hypothesized that there would not be any cost to well-being for those adolescents who reported playing a small to a moderate amount of video games, but there would be a negative impact on mental well-being for those

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adolescents who reported large amounts of video game playing. To test the hypothesis, Pryzbylski and Weinstein (2017) obtained data from a very large sample of 15-year-old adolescents who reported the amount of time they played video games per day (in hours).⁶ The adolescents also completed a 14-item self-report measure of mental well-being (happiness, life satisfaction, psychological functioning, and social functioning). I have simulated some data to correspond to the essence of the results reported by Pryzbylski and Weinstein (2017) ([Data File: video games well being](#)).

Figure C14.2. Idealised Lines of Best Fit: Four Possible Curvilinear Quadratic Effects



The key to conducting a curvilinear regression is that you need to transform your predictor variable in such a way as to reflect the nature of the curvilinear effect in which you are interested. As mentioned, a quadratic effect is by far the most commonly observed and tested curvilinear regression. A quadratic curvilinear effect implies one bend in the regression line of best fit (a linear effect implies a straight line of best fit). The bend in the line of best fit may reflect a U-shaped pattern (Figure 14.2, Panel A) or an inverted U-shaped pattern (Figure 14.2, Panel B). Often, the quadratic curvilinear pattern will not represent a perfect U-shape. Instead, for example, the line of best fit may increase across most of the first half of the scores

⁶ Pryzbylski and Weinstein (2017) also had data relevant to TV watching, computer use, and smartphone use, if you're interested.

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on the X-axis and then plateau across the second-half of the X-axis (Figure 14.2, Panel C). Finally, the opposite pattern may also be observed from time to time (Figure 14.2, Panel D).

Prior to conducting the curvilinear regression, I'll note that the Pearson correlation between hours of video-game playing and mental well-being was estimated at $r = -.116$, $p = .002$. Such a result suggests that higher levels of video game playing is associated with lower levels of mental well-being. As I show below, it would be a bit mistake to end the analysis there.

Correlations			
		gaming_hours	well_being_avg
gaming_hours	Pearson Correlation	1	-.116**
	Sig. (2-tailed)		.002
	N	734	734
well_being_avg	Pearson Correlation	-.116**	1
	Sig. (2-tailed)	.002	
	N	734	734

** . Correlation is significant at the 0.01 level (2-tailed).

The first step to conducting a quadratic curvilinear analysis is to transform the predictor variable scores into squared variables (**Watch Video 14.8: [Square a variable \(via transform\) in SPSS](#)**). Once you create the new (squared variable), you can conduct a hierarchical multiple regression analysis. In the first block, include the original independent variable. In the second block, include the squared independent variable. If the inclusion of the squared independent variable into the model is found to be associated with a statistically significant increase in model R^2 , then you can suggest that there was a curvilinear effect between the independent variable and the dependent variable. For the video game playing and mental well-being example, I obtained the following results in SPSS (**Watch Video 14.9: [Curvilinear \(Non-Linear\) Regression in SPSS](#)**).

Model Summary									
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics				
					R Square Change	F Change	df1	df2	Sig. F Change
1	.116 ^a	.014	.012	.66285	.014	10.056	1	732	.002
2	.252 ^b	.063	.061	.64633	.050	38.896	1	731	.000

a. Predictors: (Constant), gaming_hours

b. Predictors: (Constant), gaming_hours, gaming_hours2

As can be seen in the SPSS table entitled 'Model Summary', the model R and R square values associated with Model 1 (the inclusion of only video game playing as a predictor) were estimated at .116 and .014, respectively, which implies that 1.4% of the variance in the dependent variable (mental well-being) was accounted for by the independent variable (video

game playing hours). Furthermore, the model R^2 was significant statistically, $F(1, 732) = 10.06$, $p = .002$. It is not a coincidence that the model R^2 p -value was identical to the Pearson correlation p -value. They correspond to essentially the same analysis. More importantly, Model 2, which includes both the gaming hours independent variable and the squared gaming hours independent variable, yielded Model R and R^2 values of .252 and .063, respectively, which implies that 6.3% of the variance in mental well-being was accounted for by the combination of the linear (video game hours) and quadratic effects (squared video game hours). Furthermore, the R^2 change (.050; i.e., .063 - .014, within rounding) was found to be significant statistically, $F(1, 731) = 38.90$, $p < .001$. Thus, the inclusion of the squared video game hours variable increased the predictive capacity of the regression equation. Such a result implies that there is at least one bend in the line of best fit, i.e., the effect is not linear.

Prior to examining the scatter plot, you can see in the SPSS table entitled 'Coefficients' that both the gaming hours (gaming_hours) and squared gaming hours (gaming_hours2) variables were statistically significant contributors to the model, as each variable was associated with a statistically significant beta-weight: gaming_hours, $b = .182$, $\beta = .634$, $t = 5.05$, $p < .001$; gaming_hours2, $b = -.028$, $\beta = -.783$, $t = -6.237$, $p < .001$. Unfortunately, beta-weights, even standardized beta-weights, are not particularly interpretable in a curvilinear regression analysis. In the context of a quadratic curvilinear regression, the most you can really say is the following. If the quadratic effect beta-weight is positive in direction, it suggests a U-shaped pattern in the scatter plot. By contrast, if the beta-weight is negative in direction, it suggests an inverted U-shaped pattern in the scatter plot.

Some would recommend transforming the independent variable into z-scores and then square the z-scores to create a standardized variable to represent the quadratic effect in a standardized way. The argument, here, is that these standardized variables, when analyzed in the hierarchical multiple regression, will yield more interpretable beta-weights. I'm not entirely convinced of this approach, as I do not believe the standardized beta-weights become fully interpretable, in the same way that standardized beta-weights can be interpreted in a more typical (linear) multiple regression.

Instead of focusing on the beta-weights in a curvilinear linear regression, one might consider focusing on the semi-partial correlations, as they appear to be more interpretable. In this example, the quadratic term was associated with a semi-partial correlation of $r = -.223$ (see SPSS table entitled 'Coefficients'). The negative semi-partial correlation implies an inverted U-shaped effect, which is consistent with the scatter plot depicted in Figure C14.3. Furthermore, the squared semi-partial correlation of .050 (i.e., $-.223^2$) implies that 5.0% of the variance in mental well-being was accounted for by the curvilinear effect. The linear effect accounted for 1.3% of the variance in mental well-being, on the basis of the squared semi-partial correlation estimated from *model 1* (i.e., $.116^2 = .013$). I do not believe the linear effect semi-partial correlation from model 2 is interpretable in this way (so, it should probably be

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ignored). Note that the sum of the squared semi-partial correlation associated with the linear effect (model 1) and the squared semi-partial correlation associated with the quadratic effect (model 2) essentially equals the model 2 R^2 of .063 (.013 + .051 = .064). I don't know for certain if this will always be the case, but it (i.e., summing model 1 linear squared and model 2 quadratic squared) does seem to apply across several data files I have.

Before moving onto the all-important scatter plot, I'd like to point out that the collinearity statistics (Tolerance and VIF) suggest that collinearity is a problem in the data, as the tolerance is less than .10 and the VIF is greater than 10, in this example. There is much literature on the issue that the linear effect term and the quadratic effect term necessarily correlate with each very substantially (in this case, they correlated $r = .958$). The concern is that the substantial correlation will cause unacceptably high multi-collinearity. In this example, you will note that tolerance was reported at .081 and the VIF was reported at 12.29 (see 'Coefficients' SPSS table). Such values suggest a problem (recall that tolerance should not exceed .10 and VIF should be less than 10). However, in a curvilinear regression analysis, you should not trust these multicollinearity values. They are misleading and untrustworthy. If multicollinearity were really as bad as $VIF = 12.29$, the standard errors associated with the unstandardized beta weights would be larger than they are (e.g., gaming_hours2 had a tiny unstandardized standard error of .004). Furthermore, neither the linear nor the quadratic effects would be significant statistically; instead, they were both significant statistically. Again, multicollinearity is not a problem. Ignore it as an issue in regressions that include product terms, such as curvilinear regression and moderated regression (Disatnik & Sivan, 2016). Consequently, feel free to proceed with the analysis.

Coefficients ^a												
Model	Unstandardized Coefficients		Standardized Coefficients	t	Sig.	95.0% Confidence Interval for B		Correlations			Collinearity Statistics	
	B	Std. Error	Beta			Lower Bound	Upper Bound	Zero-order	Partial	Part	Tolerance	VIF
1 (Constant)	3.461	.043		79.864	.000	3.376	3.546					
gaming_hours	-.033	.011	-.116	-3.171	.002	-.054	-.013	-.116	-.116	-.116	1.000	1.000
2 (Constant)	3.199	.060		53.674	.000	3.082	3.316					
gaming_hours	.182	.036	.634	5.050	.000	.111	.253	-.116	.184	.181	.081	12.29
gaming_hours2	-.028	.004	-.783	-6.237	.000	-.037	-.019	-.175	-.225	-.223	.081	12.29

a. Dependent Variable: well_being_avg

It is almost always essential to evaluate a curvilinear regression with a scatter plot and line of best fit (**Watch Video 14.10: [Curvilinear \(Non-Linear\) Scatter Plot in SPSS](#)**). As can be seen in Figure C14.3, the scatter plot depicted an inverted U-shaped relationship between video game playing hours and mental well-being. Specifically, well-being tended to increase from about 0 hours to 3 hours of video game playing, then started to decrease beyond 3 hours

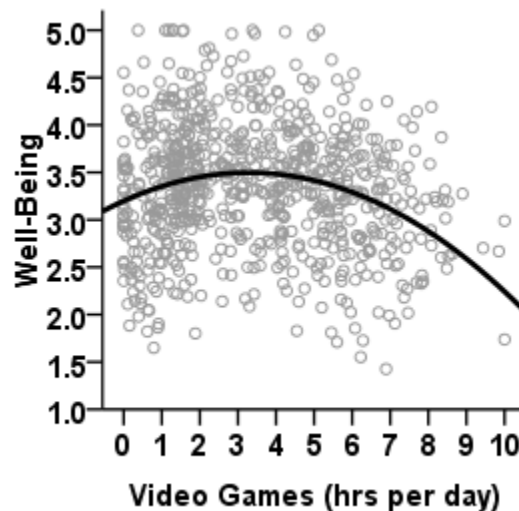
C14.30

Gignac, G. E. (2023). *How2statsbook* (Online Edition 2). Perth, Australia: Author.

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of video game playing per day.⁷ Clearly, relying only upon the Pearson correlation of $-.116$ would mislead people in a serious way, as it failed to reflect the curvilinear nature of the effect: small to moderate amounts of video game playing *appears*⁸ to be a good thing for the mental well-being of 15-year-olds!

Figure C14.3. Scatter Plot Depicting the Curvilinear (Quadratic) Association Between Video Game Playing (Hours Per Day) and Mental Well-Being



Finally, whether it is permissible to interpret a statistically significant linear effect in the presence of a statistically significant quadratic effect appears to be discussed rarely. In my view, a linear effect should not be interpreted in the presence of a statistically significant quadratic effect, as it would be misleading. Consider the current study relevant to video-game playing and mental well-being. If someone asked if video game playing was related positively or negatively to mental well-being in 15-year-olds, one could not say, categorically, ‘yes, positively’ or ‘yes, negatively’. The only way to describe the association, justifiably, is to state that the association is consistent with an inverted U-shaped effect: the association is positive from 0 to about 3 hours per day and then negative from beyond 3 hours onwards. In other words, one would have suggest that the nature and/or strength of the association depends on where along the X-axis one is considering. Recall from the factorial ANOVA chapter that when one has to state that the nature and/or strength of the effect depends on the values of

⁷ My data don’t quite replicate the results reported by Pryzbylski and Weinstein (2017). They reported peaks in well-being at 2 hours per day, for weekdays, and 3 hours per day, for the weekends.

⁸ Based on these results, one could not contend justifiably that playing a small to moderate amount of video games per day *causes* an increase in mental well-being. These data are non-experimental in nature, consequently, one could only suggest the possibility of a causal effect. When I was 15, I played about 1 hour per day on my commodore 128 (in commodore 64 mode). Those were the days!

another variable, one is dealing with an interaction. Correspondingly, you may regard a curvilinear regression as the most basic type of interaction in a regression context: the nature and/or strength of the association between X and Y depends on the values of X . In the next section of this chapter, I describe moderated (interaction) multiple regression with two different independent variables and one dependent variable.

Moderated Multiple Regression

We probably all know a few people with a large amount of motivation to succeed in life, but not much in the way of brains, and, so, they achieve a certain amount in life, but not really an impressive amount. Conversely, we probably all know a few people with a high level of intelligence, but little to no motivation to succeed, and they achieve very little over the years. Research has shown that both intelligence and motivation are ingredients for success (e.g., Duckworth et al., 2011). However, some have theorized that, when you combine elevated levels of intelligence and elevated levels of motivation in the same person, something special happens: on average, you observe a disproportionate amount of success from this type of person. That is, more so than you would predict from their intelligence and motivation levels added-up as individual predictors. Instead, the combination of the two variables, intelligence and motivation, has been theorized to have a multiplicative (“synergistic”) effect on success. Anyway, that’s theory. It has been difficult to find empirical evidence for this theory. In the area of academic success, However, Hirschfeld, Lawson and Mossholder (2004) argued that it was important to measure motivation in a context-specific way, not general motivation. Consequently, they obtained scores of general intelligence from 364 undergraduate university students. They also had the students complete a short academic achievement motivation self-report questionnaire (e.g., ‘I always strive to achieve excellence in my school work.’). Finally, the dependent variable was grade point average (GPA). Thus, each of the three variables was measured on a continuous scale.

Prior to getting on with the analysis, I’ll note that, conceptually, you should think of moderated multiple regression as similar to interactions in the context of factorial ANOVA. Recall that an interaction was described as consistent with the observation of a difference in the magnitude of the differences between means across levels of an independent variable. The main difference is that in ANOVA, the independent variables are measured on a nominal scale, whereas in multiple regression the independent variables are measured on a continuous (ratio/interval/ordinal) scale. Of course, the dependent variable is measured on a continuous scale for both types of analyses.

From a procedural perspective, the key element to conducting a moderated multiple regression is the requirement to multiply the two independent variables together. Such a process creates a product term (the multiplication of two scores yields a ‘product’). The

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product represents the interaction term in the analysis. As per a curvilinear analysis, one can test a hypothesized interaction effect via a hierarchical multiple regression. At step 1, the two independent variables are entered into the equation. Then, at step 2, the product variable is entered into the equation. If a statistically significant R^2 change (as tested via F -change) is observed between step 1 and step 2, a statistically significant interaction may be declared.

I simulated some data to correspond to the results reported by Hirschfeld et al. (2004) ([Data File: IQ_motivation_gpa](#)). First, I created the product term by multiplying the intelligence scores by the academic achievement scores. In SPSS this can be done via the 'Compute variable' utility (**Watch Video 14.11: [Create Product Term Variable in SPSS](#)**). Next, I examined the Pearson correlations between the variables, which is always good practice for virtually any multiple regression analysis.

		GPA	intelligence	academic motivation	intelligence_by _motivation
GPA	Pearson Correlation	1	.420**	.350**	.512**
	Sig. (2-tailed)		.000	.000	.000
	N	364	364	364	364
intelligence	Pearson Correlation	.420**	1	.060	.559**
	Sig. (2-tailed)	.000		.254	.000
	N	364	364	364	364
academic motivation	Pearson Correlation	.350**	.060	1	.856**
	Sig. (2-tailed)	.000	.254		.000
	N	364	364	364	364
intelligence_by_motivation	Pearson Correlation	.512**	.559**	.856**	1
	Sig. (2-tailed)	.000	.000	.000	
	N	364	364	364	364

** . Correlation is significant at the 0.01 level (2-tailed).

As can be seen in the SPSS table entitled 'Correlations', both intelligence ($r = .42$, $p < .001$) and academic motivation ($r = .35$, $p < .001$) correlated positively with academic performance (GPA). However, intelligence and academic motivation did not correlate with each other significantly ($.06$, $p = .254$). Furthermore, the intelligence by motivation product term correlated positively with GPA ($r = .51$, $p < .001$). Finally, the product term correlated with intelligence and academic achievement at .56 and .86. Am I worried about the large correlation of .86 between the product term and academic motivation? Not one bit. As far as multicollinearity is concerned, it is not a problem. It's an illusion.

Next, I conducted the hierarchical multiple regression in SPSS (**Watch Video 14.12: [Moderated Multiple Regression in SPSS](#)**). There are three key tables to consult to interpret the results fully. The 'Model Summary' table, the 'ANOVA' table, and the 'Coefficients' table. As can be seen in the 'Model Summary' table below, the Model 1 results are reported, which includes the model R and R square associated with the two predictors included in the analysis.

Thus, 28.2% of the variance in GPA was accounted for by intelligence and academic achievement, which was significant statistically, $F(2, 361) = 70.99, p < .001$. We don't know which variable contributed to the equation, yet. Those results are reported in a table 'Coefficients' table, where it can be seen that both intelligence and academic motivation were statistically significant contributors to the model. Specifically, intelligence: $b = .002, \beta = .40, t = 8.97, p < .001$; academic achievement: $b = .24, \beta = .33, t = 7.30, p < .001$.

The most important piece of information in the SPSS 'Model Summary' table is the 'R square change' and associated F - and p -value. In this case, the R square change was equal to .010, which was statistically significant, $F(1, 360) = 4.97, p = .026$. The statistically significant R square change implies that the moderator hypothesis was supported: it does appear to be the case that the association between intelligence and GPA interacts with academic motivation. One could just as appropriately state that the association between academic motivation and GPA interacts with intelligence. They are two sides of the same coin. As can be seen in the SPSS table entitled 'Coefficients', the intelligence by academic motivation product term was associated with a statistically significant beta-weight, $b = .001, \beta = 1.06, t = 2.23, p = .026$. The overall model including all these independent variables (i.e., Model 2) was associated with a model R^2 .292 (model $R = .540$), which was significant statistically, $F = 49.50, p < .001$. It's not a coincidence that the sum of squares 'Regression' divided by the sum of squares 'Total' is equal to the model R^2 value (i.e., $30.913 / 105.851 = .292$). As I describe further below, there is a close correspondence between ANOVA and multiple regression. SPSS makes use of an ANOVA framework to test the final model R^2 for statistical significance, hence the inclusion of a table entitled 'ANOVA'.

Interpreting a statistically significant moderator effect in multiple regression is not easy, especially if all you rely upon is the beta-weight. As per curvilinear regression, a positively directed interaction term beta-weight implies that the association between the independent variable and the dependent variable becomes progressively stronger across the spectrum of the moderator variable values. In this study, which variable you would consider the independent variable and which variable the moderator is arbitrary. Theoretically and empirically, it is not possible to assert whether intelligence is the moderator of the effect between academic achievement and GPA, or whether academic achievement is the moderator of the effect between intelligence and GPA. It's best to think about the moderator effect as a mutual, multiplicative (synergistic) effect.

Prior to demonstrating a graphical method to depict the nature of a moderated effect, it is useful to point out that neither the unstandardized nor the standardized beta-weights associated with the interaction term are readily interpretable. For example, in this example investigation, the standardized beta-weight associated with the interaction term exceeded 1.0, which is possible, mathematically, it's just not easy to interpret in an insightful way. Instead, I recommend that you focus upon the semi-partial correlations associated with a moderated

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multiple regression analysis, as per my recommendation for a curvilinear regression. That is, by summing the squared semi-partial correlations associated with the main effects included in model 1 (i.e., $.400^2 + .325^2$) and the product term squared semi-partial correlation reported in model 2 (i.e., $.10^2$), you get close to the overall model 2 *R* square of .292 (i.e., $.160 + .106 + .010 = .276$). The difference between the two *R* square values, $.292 - .276 = .016$, represents the percentage of variance accounted for in the dependent variable due to the shared variance between the predictor variables. Recall that intelligence and motivation were inter-correlated at only $r = .060$, so, it makes sense that the common variance predictive influence on the dependent variable was fairly small, in this example. I also note, briefly, that presence of a suppressor effect would complicate the interpretation of the difference between the .292 and .276 *R* square values. I discuss suppressor effects further in this chapter.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics				
					R Square Change	F Change	df1	df2	Sig. F Change
1	.531 ^a	.282	.278	.45875	.282	70.989	2	361	.000
2	.540 ^b	.292	.286	.45625	.010	4.966	1	360	.026

a. Predictors: (Constant), academic motivation, intelligence

b. Predictors: (Constant), academic motivation, intelligence, intelligence_by_motivation

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	29.879	2	14.940	70.989	.000 ^b
	Residual	75.972	361	.210		
	Total	105.851	363			
2	Regression	30.913	3	10.304	49.502	.000 ^c
	Residual	74.938	360	.208		
	Total	105.851	363			

a. Dependent Variable: GPA

b. Predictors: (Constant), academic motivation, intelligence

c. Predictors: (Constant), academic motivation, intelligence, intelligence_by_motivation

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Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Correlations			Collinearity Statistics	
		B	Std. Error	Beta			Zero-order	Partial	Part	Tolerance	VIF
1	(Constant)	-.105	.244		-.432	.666					
	intelligence	.002	.000	.400	8.965	.000	.420	.427	.400	.996	1.004
	academic motivation	.244	.034	.326	7.297	.000	.350	.359	.325	.996	1.004
2	(Constant)	2.450	1.172		2.091	.037					
	intelligence	-.001	.001	-.138	-.562	.575	.420	-.030	-.02	.033	30.69
	academic motivation	-.409	.295	-.545	-1.386	.167	.350	-.073	-.06	.013	78.74
	intelligence by motivation	.001	.000	1.056	2.228	.026	.512	.117	.10	.009	114.2

a. Dependent Variable: GPA

An arguably more sophisticated, but more difficult to understand, approach to evaluating the relative influence of predictors in a moderated multiple regression is a commonality analysis (a.k.a., element analysis). Commonality analysis decomposes the unique and common percentages of variance accounted for by each individual variable, as well as all possible combinations of the variables, included in the model. When there are three variables entered into the multiple regression analysis, there are 7 possible variable combinations that may have accounted for a proportion of the explained variance attributable to the regression equation. In this example study, 29.2% of the variance in GPA was accounted for by the multiple regression equation. However, it is not known how that percentage of variance arose. Unfortunately, it is not as simple as summing the squared semi-partial correlations associated with each variable included in the analysis, as the inter-predictor variable correlations would imply that some of the overall R^2 arose due to combinations of the predictor variables, hence the term 'commonality analysis'. Fortunately, Nimon (2010) developed a script that can be run within SPSS to conduct a commonality regression analysis, which I applied to the current intelligence, academic achievement and GPA data. As can be seen in Table C14.2 (left-side), the 29.2% of the variance shared between the predictors (IQ and motivation) and the dependent variable was not much due to the unique effects of IQ, motivation, or the moderator term. Instead, the vast majority of the shared variance was due to the common IQ and IQ*motivation effect (54.50%), as well as the common motivation and IQ*motivation effect (34.96%). Only a relatively small percentage (8.68%) of the shared variance was due to the common IQ and motivation effect.

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Table C14.2. Results of Commonality Regression Analysis: IQ, Motivation, GPA

Individual Combinations			Individual Variables			
Variables	Coefficient	Percent	Variable	Unique	Common	Total
Unique to iq	.0006	.2125	iq	.0006	.1758	.1764
Unique to mot	.0038	1.2932	mot	.0038	.1187	.1225
Unique to iqmot	.0098	3.3440	iqmot	.0098	.2526	.2623
Common to iq mot	.0253	8.6671				
Common to iq iqmot	.1592	54.4975				
Common to mot iqmot	.1021	34.9606				
Common to iq mot iqmot	-.0087	-2.9748				
Total	.2920	100.0000				

Moderation: Spotlight Analysis and Plots

In addition to a commonality regression analysis, it is recommendable to create a visual representation of a moderator effect. The essential strategy associated with depicting a moderated regression effect is to represent the association between the independent variable and the dependent variable across two or more levels of the moderator variable. Such a strategy may be considered consistent with a 'spotlight test' (Spiller, Fitzsimons, Lynch, & McClelland, 2013), which may incorporate the estimation of the correlation between the independent and dependent variables across the different levels of the moderator. When there is a statistically significant moderator effect, the magnitude/direction of the correlations will be unequal.

In practice, given that a moderator is often measured on a continuous scale in a typical moderated multiple regression, the spotlight analysis levels need to be created artificially. For example, one may split the sample into two groups: low-moderate versus moderate-high. In practice, applying such a strategy would involve calculating the median value on the moderator and giving all cases with a value less than and equal to the median on the moderator a '1' and all cases with a value greater than the median on the moderator a value of '2'. Then, create two scatter plots with the inclusion of a line of best fit, as well as the correlation for each of the two portions of the sample.

I conducted such a spotlight analysis with the IQ, motivation and GPA data (**Watch Video 14.13: [Spotlight Analysis in SPSS - Basic](#)**). As can be seen in Figure C14.4, the two scatter plots depicting the association between intelligence and GPA across the two levels of academic motivation (low to moderate versus moderate to high) were a little different. Specifically, in the low to moderate portion of the sample, the angle of the slope was slightly lower than the angle of the slope for the moderate to high academic motivation group. Correspondingly, the Pearson correlation between intelligence and GPA was larger ($r = .46$) for the moderate to high portion of the sample, in comparison to the corresponding Pearson correlation for the low to moderate motivation portion of the sample ($r = .38$). The statistically

C14.37

significant interaction detected by the moderated multiple regression (R^2 change = .010, $F = 4.97$, $p = .026$), implies, at least in rough terms, that the two correlations are statistically significantly different from each other. However, this will not necessarily be the case, as there are many ways to split-up the sample across the moderator continuum. In this example, the statistically significant interaction implies that the correlation between intelligence and GPA was not equal across all levels of the academic motivation. In practice, we do not know exactly where the split-sample correlations are unequal across the spectrum of the moderator. Splitting the sample into low and high groups is good start, and my demonstration above will hopefully give you some insight into the graphical spotlight analysis.

Fortunately, rather than estimate the median and create the two groups manually, O'Connor (1998) created a nice, little SPSS syntax file that will conduct a 'split slope' analysis in a relatively efficient way (**Watch Video 14.14: [Spotlight Analysis in SPSS - Extreme](#)**). The default in the O'Connor (1998) syntax file is to estimate the independent variable and dependent variable slopes across three levels of the moderator: (1) one standard deviation below the moderator mean, (2) the moderator mean, and (3) one standard deviation above the moderator mean. Additionally, the X-axis data points are generated by the syntax file on the basis of one standard deviation below and above the independent variable mean. When I ran the O'Connor (1998) syntax with the default settings, I obtained the chart depicted in Figure C14.5 (left-side). It can be seen that the degree of positive slope increased across the three levels of academic motivation. Thus, the magnitude of the association between intelligence and academic achievement (GPA) increased across the continuum of academic motivation. Thus, the combination of intelligence and academic motivation impacts academic performance in such a way that we would not predict, based only on an examination of the intelligence and motivation scores, separately. Something a little "special" appears to happen when relatively elevated levels of intelligence are combined with relatively elevated levels of academic motivation.

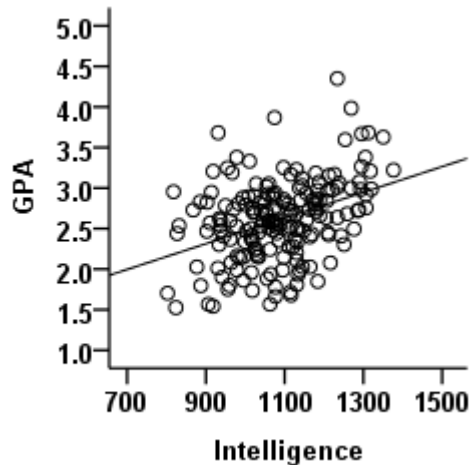
As mentioned, the default setting in the O'Connor (1998) syntax is to split the moderator data across three groups, two of which are one standard deviation above and below the mean (get syntax [here](#)). You can manipulate the O'Connor (1998) syntax such that the moderator data are split across three groups, two of which are two (rather than one) standard deviations above and below the mean. Increasing the standard deviation from one to two standard deviations has the effect of accentuating the magnitude of the interaction effect. As can be seen in Figure C14.5 (right side), the difference in the degree of slope is more substantial for two standard deviation criterion, in comparison to the one standard deviation criterion. Consequently, in my view, the split slopes analysis commonly used by researchers tends to exaggerate the magnitude of the interaction effect. By comparison, my original split slopes analysis based on low-moderate and moderate high groups (Figure C14.4) is arguably a more appropriate (i.e., realistic) method to represent a two-way interaction for a moderator

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analysis (you can also present a single scatter plot with two lines of best fit, as I demonstrate in the video), as it is more consistent with the moderator multiple regression results, which is based on all of the data (not extreme groups data). Finally, you might find it beneficial to split the sample into a larger number of finer portions for a more comprehensive and refined ‘floodlight analysis’ (see Spiller et al., 2013).

Figure C14.4. Scatter Plots for Split Slope Analysis (Raw Data)

Low to Moderate Motivation ($r = .38$)



Moderate to High Motivation ($r = .46$)

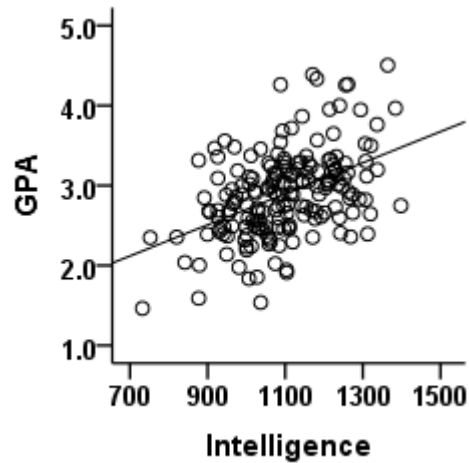
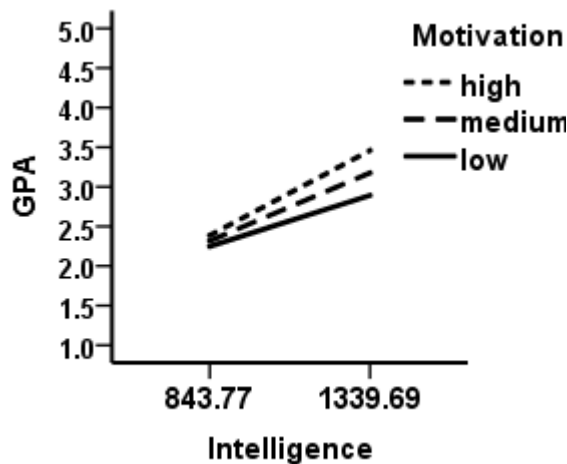
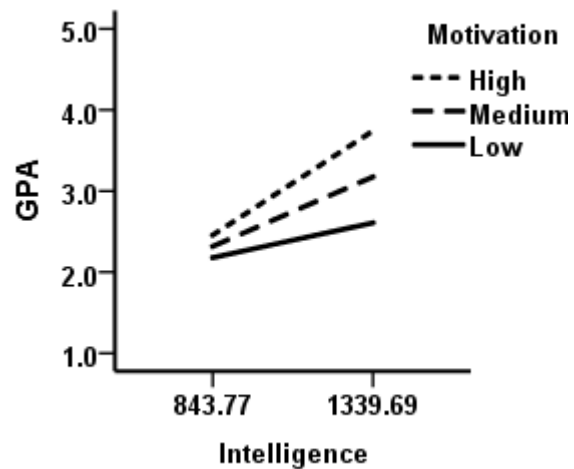


Figure C14.5. Scatter Plot

Moderator: 1 SD Above/Below Mean



Moderator: 2 SD Above/Below Mean



Multicollinearity in Moderated Regression

At face value, multicollinearity should be considered a problem in moderated multiple regression, as one or both of the predictors in a moderated multiple regression will necessarily correlate very highly ($.80+$) with the product term derived from the two independent variables.

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Correspondingly, the tolerance and variance inflation factor indicators will almost always suggest that multicollinearity is grossly excessive. Consequently, researchers tend to apply strategies to reduce the degree of shared variance between the predictors and the product term. One commonly observed strategy involves subtracting each predictor variable's mean from the case values for a given variable, prior to calculating the product term to represent the moderator effect. Such a strategy is known as 'mean centering'.

Although mean centering will often reduce multicollinearity to some degree, there is much debate on whether multicollinearity is truly a problem for moderated multiple regression. Personally, I'm in the 'it's not a problem camp'. Like others, I believe the perceived multicollinearity is really just an illusion. As demonstrated by Disatnik and Sivan (2016), for example, the multicollinearity apparent when a product term is included in a multiple regression is due to interval scaling. Furthermore, despite the observation of a very large variance inflation factor in a moderated regression, multicollinearity due to interval scaling does not impact (increase) standard errors for the beta-weights (Disatnik & Sivan, 2016). Again, this sort of multicollinearity is an illusion. It should be kept in mind, however, that multicollinearity due to a substantial amount of shared variance between separately measured independent variables (i.e., not product terms) should be evaluated and taken into account, when it is observed.

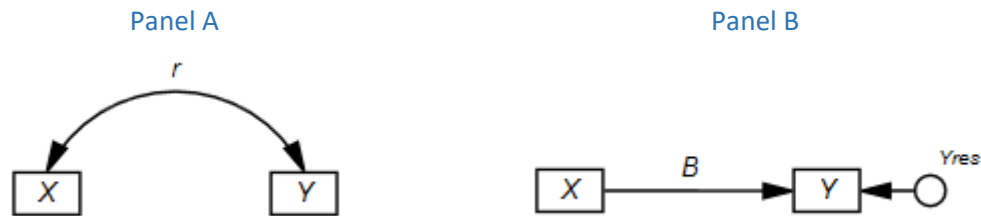
Mediated Multiple Regression

The association between two continuously scored variables may be estimated via a Pearson correlation or a bivariate regression (standardized) coefficient. As you may know, both analyses yield the same results. Although a Pearson correlation and a bivariate regression (standardized) will yield the same result, they can be considered, theoretically, to represent two different types of associations. With respect to the Pearson correlation, there is no suggestion of causality: the two variables are simply suggested to be associated. By contrast, a bivariate regression result may be interpreted to imply that the independent variable causes, at least to some degree, the dependent variable. Importantly, there is no way to know whether the independent variable causes the dependent variable with a bivariate regression. I'm only saying, here, that a researcher may imply the possibility of a causal effect between X and Y in the context of a bivariate regression. Given the different approaches to interpreting the nature of a Pearson correlation and a bivariate regression, they can also be represented visually in their respective ways. As can be seen in Figure C14.6 (Panel A), a Pearson correlation is represented by a curved arrow between two squares. One of the squares represents the independent variable (X) and one variable represents the dependent variable (Y). By comparison, a bivariate regression is represented by a straight arrow between the two squares, where the arrow leads from the independent variable to the dependent variable

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(Figure 14.6, Panel B). In the context of mediation analyses, a bivariate regression coefficient (B) between X and Y is often referred to as a **total effect**. The direction of the arrow may be considered to imply that the independent variable causes the dependent variable. Again, there is no way to establish causality with a bivariate regression. However, more complicated models can be tested to help suggest more convincingly the possibility of causality.

Figure C14.6 Visual Depictions of the Pearson Correlation and Bivariate Regression Models



One such model is a mediation regression model. In its simplest form, the purpose of mediated regression is to help understand the nature of an association between two variables. Specifically, a researcher may hypothesize that the total effect between two variables (X and Y) may be due, partially or fully, to a third variable. A variable hypothesized to account, partially or fully, the association between two variables is known as a mediator (M). When a variable is found to account, partially or fully, the association between X and Y , researchers will often state that there is an **indirect effect** between X on Y via M . In practice, the generation of a hypothesis that includes a mediator should be supported by theoretical considerations.

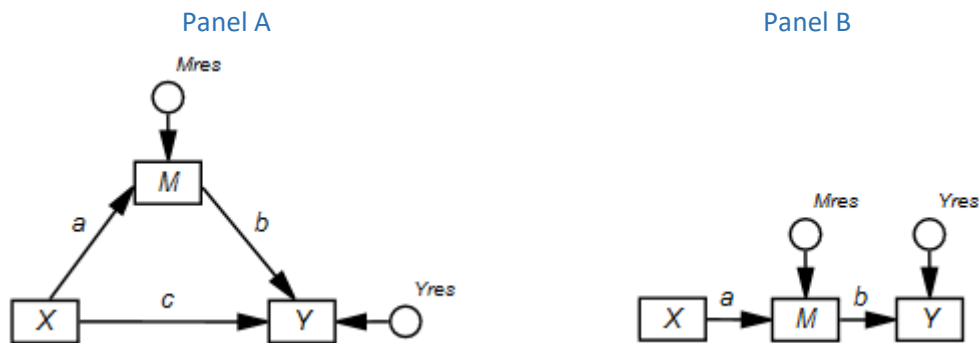
For example, Stoeber and Corr (2016) were interested in the association between self-oriented perfectionism and positive affect. As the name implies, self-oriented perfectionism pertains to having high expectations for oneself, rather than the sort of perfectionism that is perceived by a person to be imposed on them by others. Theoretically, one may hypothesize that self-oriented perfectionism (X) influences individual differences in state positive affect (Y) (feelings of enthusiasm, pride, etc.). Furthermore, it may be hypothesized that the association between self-oriented perfectionism and positive affect may be due (mediated), partially or fully, to individual differences in flourishing (M) (self-perceived success).

The hypothesized mediation models are depicted visually in Figure C14.7 (**Watch Video 14.15: [Mediation - Purpose & Key Terms](#)**). In the partial mediation model (Panel A), there is a straight arrow leading from X to Y (c). Additionally, there is an arrow leading from X to M (a) and an arrow leading from M to Y (b). The coefficient associated with path (a) is simply a bivariate regression coefficient. Paths (b) and (c) represent multiple regression beta weights. Prior to testing the model depicted in Panel A, one would expect to see a statistically significant total effect between self-oriented perfectionism and positive affect (based on a Pearson correlation or a bivariate regression). When partial mediation is observed, the direct

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effect (c) between X and Y will reduce in magnitude, but remain significant statistically ($p < .05$). Thus, the observation of partial mediation will involve three direct effects that are statistically significant (a , b , and c ; see Figure C14.7, Panel A). In addition to the three direct effects, there is also one indirect effect associated with the partial mediation model, but you can't see it obviously in the model. The indirect effect corresponds to the product of the $X \rightarrow M$ (a) and $M \rightarrow Y$ (b) direct effect coefficients. Stated alternatively, the indirect effect represents the association between X and Y that is due to the influence of M . In practice, testing the statistical significance of the indirect effect is the tricky part, as most statistical programs do not test them in an automatic way. However, in order for partial (or full) mediation to be supported, the indirect effect must be found to be significant statistically.

Figure C14.7 Visual Depictions of Partial (Panel A) and Full (Panel B) Mediation Models



In contrast to partial mediation, full mediation is observed when there is a significant total effect between X and Y (c). However, once the mediator is added to the model, the direct effect between X and Y (c) is not found to be significant statistically. Additionally, the indirect effect between X and Y ($a*b$) must be observed to be significant statistically to support full mediation. The direct effect (c) does not have to be exactly zero to suggest full mediation. The direct effect coefficient just has to be non-significant statistically (with a corresponding significant indirect effect). When full mediation is observed, one could represent the nature of the associations between the variables as per Figure C14.7 (Panel B), because the model implies that there is no direct effect between X and Y . That is, the entire total effect between X and Y is presumed to be due to the mediator. Consequently, X cannot be considered a cause of Y . Stated alternatively, one might suggest the possibility that the association between X and Y is caused by M .

To demonstrate a mediation analysis with one hypothesized mediator, I return to the Stoeber and Corr (2016) study, where the association between self-oriented perfectionism and positive affect is hypothesized to be mediated, partially or fully, by individual differences in flourishing ([Data File: perfectionism flourishing](#)). First, I estimated the bivariate regression coefficient between self-oriented perfectionism and positive affect (**Watch Video 14.16:**

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[Testing Mediation in SPSS - Example 1](#)). As can be seen in the SPSS table entitled 'Coefficients', the unstandardized (.102) and standardized (.140) beta-weights were significant statistically, $p = .006$. Thus, there was a total effect to potentially mediate.

Coefficients^a

Model	Unstandardized Coefficients		Standardized Coefficients	t	Sig.
	B	Std. Error	Beta		
1	(Constant)	2.690	.174	15.477	.000
	self_perf	.102	.037	2.771	.006

a. Dependent Variable: pa

Next, I regressed the hypothesized mediator (flourishing) onto the self-oriented perfectionism variable, in order to estimate the coefficient (a) depicted in Panel A of Figure C14.7. As can be seen in the SPSS table entitled 'Coefficients' below, the effect was positive and significant statistically: unstandardized beta-weight = .197; standardized beta-weight = .190; both $p < .001$.

Coefficients^a

Model	Unstandardized Coefficients		Standardized Coefficients	t	Sig.
	B	Std. Error	Beta		
1	(Constant)	3.926	.247	15.905	.000
	self_perf	.197	.052	3.792	.000

a. Dependent Variable: flourish

Next, I submitted the three variables to a multiple regression analysis, in order to estimate the (b) and (c) coefficients depicted in the Panel A model of Figure C14.7. As can be seen in the SPSS table below entitled 'Coefficients', the self-oriented perfectionism was not observed to be associated with a statistically significant direct effect onto positive affect, unstandardized beta weight = .017, standardized beta-weight = .023, $p = .573$. However, the unique effect (b) of flourishing onto positive affect was significant statistically: unstandardized beta weight = .430, standardized beta-weight = .616, $p < .001$. The fact that the total effect between self-oriented perfectionism and positive effect was significant statistically, but the corresponding direct effect with the inclusion of flourishing as a mediator was not, suggests that fully mediation may have occurred. To ensure that full mediation can be interpreted convincingly, it is necessary to demonstrate that the indirect effect between self-oriented perfectionism and positive affect was significant statistically. Prior to testing the indirect effect for statistical significance, I will note that the unstandardized indirect effect ($a*b$) point-estimate corresponded to .085 (.197*.430). Correspondingly, the standardized indirect effect point-estimate corresponded to .117 (.190*.616).

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Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	1.003	.178		5.648	.000
	self_perf	.017	.030	.023	.564	.573
	flourish	.430	.028	.616	15.082	.000

a. Dependent Variable: pa

Unfortunately, SPSS does not provide a test of the indirect effect in an automatic way. Fortunately, there are SPSS syntax/macro files that can be used to test an indirect effect for statistical significance. A particular popular approach was published by Preacher and Hayes (2004). You can download the syntax file here: <https://tinyurl.com/y9h3bqlx>. Preacher and Hayes' (2004) program can test the significance of an indirect effect in two ways: (1) Sobel test; and (2) bootstrapping. With respect to the Sobel test, I obtained the following key results from the Preacher and Hayes (2004) SPSS macro:

INDIRECT EFFECT AND SIGNIFICANCE USING NORMAL DISTRIBUTION

	Value	s.e.	LL95CI	UL95CI	Z	Sig(two)
Effect	.0849	.0231	.0395	.1302	3.6702	.0002

It can be seen that unstandardized indirect effect was estimated at .0849 (same as I calculated by hand). Furthermore, the standard error was .0231, which yielded a z-value of 3.6702 (i.e., .0849/.0231) and a *p*-value of .0002. Thus, the indirect effect between self-oriented perfectionism and positive affect via flourishing was significant statistically. Technically, these results correspond only to the unstandardized effects. You would only be able to imply that the standardized indirect effect (i.e., .117) was also significant statistically, based on the Sobel test.

Fortunately, Preacher and Hayes' macro (2004) also provides the option to test the statistical significance of an indirect effect via bootstrapping. Some arguments have been made that the Sobel test does not provide accurate results in all circumstances, particularly when the sample size is small. Additionally, there is reason to believe that the sampling distribution of the product of two coefficients may not be normal. Consequently, Preacher and Hayes (2004) recommend testing the statistical significance of indirect effect via bootstrapping. As you may know, already, bootstrapping does not assume any level of normality. When I applied the Preacher and Hayes (2004) macro based on 2,000 bootstrapped samples, I obtained the following results.

BOOTSTRAP RESULTS FOR INDIRECT EFFECT

	Data	Mean	s.e.	LL99 CI	LL95CI	UL95CI	UL99CI
Effect	.0849	.0844	.0236	.0251	.0400	.1333	.1536

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It can be seen that the unstandardized indirect effect standard error is only slightly different, in this case. Furthermore, the 95% confidence intervals associated with the point-estimate (.0849) did not intersect with zero, which is consistent with a statistically significant indirect effect. Mediation models with only one hypothesized mediator are known as ‘simple’ mediation models. Often, a researcher may hypothesize the possibility of two or more mediators in the same model. In my opinion, with such relatively complicated models, I would encourage researchers to move onto a path analytic framework, which can be employed with a program such as Amos, for example. If you do not have access to a path analysis program, you may also consider Hayes’ more comprehensive mediation testing program known as PROCESS (a google search should find the most up-to-date version).

To provide an example of partial mediation, I have conducted an additional mediation analysis on the following variables included in the Stoeber and Corr (2016) study: social-oriented perfectionism (a bad thing; perfectionism perceived to be imposed by others), flourishing, and negative affect. In this case, one may hypothesize that a negative association (total effect) may be present between socially-oriented perfectionism and negative affect. Furthermore, flourishing may mediate, partially or fully, the total effect between socially-oriented perfectionism and negative affect. First, I estimated the total effect between X and Y , which was found to be significant statistically, unstandardized beta-weight = .390, standardized beta-weight = .430, $p < .001$. Thus, there was the possibility of observing mediation.

Coefficients^a

Model	Unstandardized Coefficients		Standardized Coefficients	t	Sig.
	B	Std. Error	Beta		
1 (Constant)	.870	.163		5.352	.000
social_perf	.390	.042	.430	9.333	.000

a. Dependent Variable: na

Next, the beta-weight between the independent variable (X) and the mediator (M) was estimated (i.e., coefficient a), which was also significant statistically: unstandardized beta-weight = -.299; standardized beta-weight = -.240, $p < .001$.

Coefficients^a

Model	Unstandardized Coefficients		Standardized Coefficients	t	Sig.
	B	Std. Error	Beta		
1 (Constant)	5.977	.241		24.849	.000
social_perf	-.299	.062	-.240	-4.845	.000

a. Dependent Variable: flourish

Finally, a multiple regression was performed by regressing the negative affect dependent variable (Y) onto the social-perfectionism independent variable (X) and the flourishing

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mediator variable (*M*). As can be seen in the SPSS table entitled 'Coefficients', the flourishing mediator was associated with negative affect significantly: unstandardized beta-weight = -.167; standardized beta-weight = -.230, $p < .001$. Importantly, there was also a statistically significant direct effect between social-perfectionism and negative affect, unstandardized beta-weight = -.167 and standardized beta-weight = -.230, $p < .001$.

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	1.869	.255		7.339	.000
	social_perf	.340	.042	.375	8.140	.000
	flourish	-.167	.033	-.230	-4.996	.000

a. Dependent Variable: na

Thus, at best, partially mediation may be declared. In order to support a partial mediation interpretation, the indirect effect between social-perfectionism (*X*) and negative affect (*Y*), which, in unstandardized form, corresponded to .050 ($-.299 \times -.167$), needs to be tested for statistical significance. As per the full mediation result above, I tested the indirect effect for significance with the Preacher and Hayes (2004) SPSS macro (**Watch Video 14.17: Testing Mediation in SPSS - Example 2**). As can be seen below, the indirect effect was statistically significant via the Sobel test, $z = 3.44$, $p = .0006$, as well as the bootstrapped estimation approach, 95%CI: .02/.08. Thus, as the direct effect between social-perfectionism and negative affect was significant and the indirect effect between social-perfectionism and negative affect was statistically significant, it may be suggested that the partial mediation hypothesis was supported. From a psychological perspective, social-perfectionism appears to be a pretty bad phenomenon, as it has a direct effect on negative affect, beyond the indirect effect via a negative influence on flourishing.

INDIRECT EFFECT AND SIGNIFICANCE USING NORMAL DISTRIBUTION

	Value	s.e.	LL95CI	UL95CI	Z	Sig(two)
Effect	.0500	.0145	.0215	.0785	3.4427	.0006

BOOTSTRAP RESULTS FOR INDIRECT EFFECT

	Data	Mean	s.e.	LL99 CI	LL95CI	UL95CI	UL99CI
Effect	.0500	.0502	.0153	.0180	.0232	.0843	.0939

Does the total effect between *X* and *Y* need to be significant?

If the total effect between *X* and *Y* is not significant statistically in the first instance (Figure C14.6, Panel B), there is no possibility of observing partial or total mediation: there is

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nothing to mediate. As I describe in the Advanced Topics section of this chapter devoted to suppressor effects, however, it is sometimes useful to test the possibility that a total effect will increase in magnitude by the inclusion of one or more other variables, even if the total effect is near zero! Such an analysis is not a mediation analysis. It is a suppression analysis.

Mediation versions Moderation

Although mediated multiple regression is distinctly different to moderated multiple regression, I know of no other two statistical analyses that are confused more commonly, even amongst researchers with decades of research experience. Obviously, there is something about these two concepts that are difficult to distinguish for most people. My tip to you is to always keep in mind that the concept of moderation is identical to the concept of interaction in the context of ANOVA. So, in for both moderation and interaction, the magnitude and/or direction of an effect depends upon the levels of another variable. A moderator is an interacting variable.

In contrast to moderation, mediation is about testing the possibility that one or more ‘third’ variables may account for the effect between X and Y , either partially or fully. The notion of an indirect effect is unique to mediation. There are no indirect effects in a moderation analysis. Mediation within the context of multiple regression is connected to the ANOVA framework through ANCOVA. That is, the difference between two means may be reduced (partial mediation) or reduced entirely (full mediation) in an ANCOVA. In the Advanced Topics section of this chapter, I show how ANCOVA and partial correlation are exactly the same analyses.

Advanced Topics

Proving that Multiple R is Really Just a Pearson r

To help understand the nature of multiple R , it is a good exercise to calculate the predicted values of Y and correlate those values with the actual Y values. In the context of the education, IQ and earnings example, the unstandardized regression equation can be used to estimate each case's predicted Y value:

$$y = -56261.421 + 3756.289X_1 + 501.978X_2$$

where X_1 = years of education completed, and X_2 = IQ. To calculate each case's predicted annual earnings value efficiently in SPSS, you can use the Transform -> Computer Variable utility, or you can use the 'Save' utility.

If you run a Pearson correlation between the predicted values and the annual earnings values, you should get a correlation of $r = .361$, which corresponds exactly to the multiple R value of .361 obtained from the multiple regression analysis. Consequently, multiple R may be regarded as a Pearson correlation. Historically, some people have made the distinction between "multiple regression" and "multiple correlation". Multiple regression refers to all of the values relevant to the intercepts and beta weights. Multiple correlation refers to all of the values relevant to the model R and model R^2 values (**Watch Video 14.18:** [Multiple \$R\$ and Pearson's \$r\$ - The Same?](#)).

Multiple Regression: Bootstrapping

In the Foundations section of the textbook, I noted that the earnings dependent variable was substantially skewed positively. However, I carried on with the normal theory multiple regression analysis for the purposes of illustration. When independent or dependent variable scores, or the residuals, are substantially skewed and/or kurtotic, an attractive and viable estimation technique is bootstrapping, as it does not assume normally distributed data, as described in the chapter devoted to Pearson correlation. In light of the non-normally distributed data, I re-performed the education, IQ and earnings multiple regression analysis in SPSS with the bootstrap estimation technique. Note that bootstrapping is an additional module in SPSS, which may have to be purchased separately. Currently, the SPSS bootstrap utility applied to a multiple regression analysis provides bootstrapped 95% confidence intervals only for the unstandardized beta-weights. If you desire 95% confidence intervals for the standardized beta-weights, I recommend you use the SPSS syntax Lorenzo-Seva, Ferrando, and Chico (2010) published to conduct a relative importance analysis, as it also includes standardized beta-weight CIs (see relative importance analysis topic later in this chapter).

When I ran the bootstrapped multiple regression, SPSS produced a table entitled 'Bootstrap for Coefficients' (**Watch Video 14.19:** [Bootstrapping Multiple Regression in SPSS](#)).

Bootstrap for Coefficients

Model	B	Bootstrap ^a				
		Bias	Std. Error	Sig. (2-tailed)	BCa 95% Confidence Interval	
					Lower	Upper
1 (Constant)	-56261.4	-265.374	20629.8	.010	-101842	-18718.0
education	3756.29	-21.137	856.712	.000	1919.96	5354.43
IQ	501.978	6.328	265.791	.060	56.367	1057.54

a. Unless otherwise noted, bootstrap results are based on 2000 bootstrap samples

It can in the table entitled 'Bootstrap for Coefficients' that education was found to be a statistically significant contributor to the model ($b = 3756.29$, $p < .001$, 95%CI: 1919.96 and 5354.43). However, IQ was not observed to be a statistically significant contributor to the model ($b = 501.98$, $p = .060$, 95%CI: 53.37 and 1057.54). Very curiously, it will be noted that the IQ variable 95% lower-bound and upper-bound confidence intervals were both positive in value, which would suggest a statistically significant effect. Thus, the non-significant p -value estimated via the bootstrap is not consistent with the confidence intervals estimated via the bootstrap. Although it is tempting to think the reason this has happened is because SPSS used two different types of bootstrapping. That is, the p -value is based on the percentile method, while the confidence intervals are estimated based on the bias corrected accelerated method (BCa). However, you will get the same (inconsistent) result if you selected the percentile 95% confidence interval approach. I suspect part of the problem might relate to the fact that these data are slightly heteroscedastic (as shown in the Foundations section of this chapter). Conventional bootstrapping can deal with non-normality, it does not deal with violation of the homoscedasticity assumption. Below, I demonstrate how to conduct a Wild bootstrap, an estimation method that can accommodate both non-normality and heteroscedasticity.

Multiple R^2 Confidence Intervals

Few (if any) statistical packages appear to provide the option to calculate confidence intervals for multiple R^2 , even though it would be useful if they did. I suspect the reason they do not is because it is relatively more complicated to calculate confidence intervals for multiple R^2 (and even r^2 from a bivariate association, for that matter). The reason it is more complicated is that squared values are one-sided. That is, there is no possibility for both positive and negative values. Consequently, you need to think about confidence intervals around point-estimates that can only take positive (or only negative) values a little differently. When a point-estimate can, theoretically, take on both positive and negative values, a non-significant effect implies that the lower-bound and upper-bound estimates straddle zero (i.e.,

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one positive and one negative). By contrast, with values such as multiple R^2 and bivariate r^2 , a non-significant effect implies that the lower-bound reaches zero (it can't be negative). To overcome this distinction, statisticians have developed methods to calculate confidence intervals for values such as R^2 and r^2 by using the non-central F -distribution (e.g., Lee, 1971). In practice, this involves more complicated calculations, in comparison to calculating confidence intervals for R or r . Fortunately, Zou (2007) developed a SPSS syntax file that can be used to calculate confidence intervals for either multiple R^2 or bivariate r^2 . You can download the syntax file here: [multiple R2 CIs](#). You can also learn how to use the syntax file by watching me use it (**Watch Video 14.20: [Multiple R2 Confidence Intervals in SPSS](#)**). When I used Zou's (2007) SPSS syntax file to calculate the confidence intervals for the model R^2 I reported for the annual earnings, education and IQ multiple regression example described in the Foundations section of this chapter (multiple $R^2 = .130$, $N = 205$, two predictors), I obtained the following results.

```
n    k  rsq  conlev  lower  upper
250  2  .1303 .9000  .06761 .19685
```

Thus, if this study were repeated many times over with different samples all sized $N = 250$, we would expect 95% of the multiple R^2 point estimates to be between .068 and .197. It is important to note that I specified 90% confidence intervals in the SPSS syntax file (i.e., conlev = .9000). That wasn't a mistake. For point-estimates that can only take-on positive values, you need to specify 90% confidence intervals to correspond to a statistical test with $\alpha = .05$.

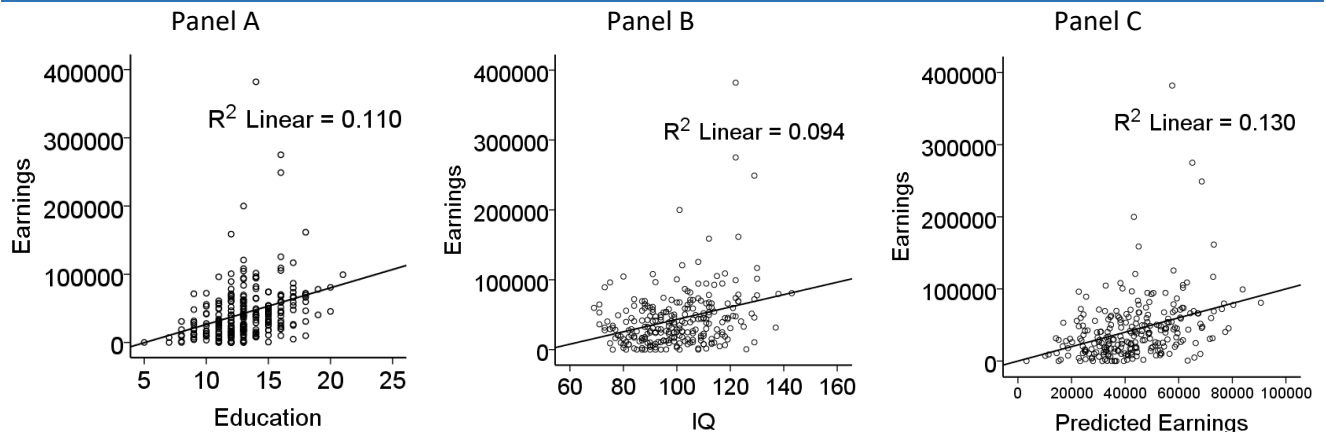
Multiple Regression Scatter Plot

Researchers often include a scatter plot in their bivariate regression reports. For example, the bivariate regression scatter plots associated with the education and IQ predictor variables with earnings as the dependent variable are depicted in Figure C14.2 (Panels A and B). In addition to bivariate regression scatter plots, it is possible to create a multiple regression scatter plot (**Watch Video 14.21: [Multiple Correlation Scatter Plot in SPSS](#)**). As can be seen in Panel C (Figure C14.8), the multiple regression scatter plot includes the predicted earnings variable on the X-axis and the actual earnings variable on the Y-axis. Across the three panels, it can be seen that R^2 was reported at .110, .094 and .130, respectively. Rarely does one ever see a multiple regression scatter plot in the literature, however, I think it can be a useful method to demonstrate the utility of the multiple regression equation. Theoretically, the multiple regression scatter plot will be associated with values more tightly centered around the fit line, in comparison to the bivariate regression scatter plots. However, in practice, there may not be much difference between the most substantial individual predictor bivariate regression scatter plot and the multiple regressions scatter plot. A lot will depend on the number of independent

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variables included in the model, as well as the magnitude of the correlations between the independent variables.

Figure C14.8. Bivariate Regression (Panels A and B) and Multiple Regression (Panel C) Scatter Plots



Testing the Difference between Two Beta Weights

I think it is plausible to suggest that a very large percentage of researchers conduct a multiple regression analysis to determine which one of the independent variables is the best unique predictor of a dependent variable. As mentioned above, it is insufficient to simply report a numerical difference between the beta weights as meaningful statistically. Instead, the numerical difference between two beta weights must be tested statistically.

One method to test the difference between two standardized beta weights is to estimate the 95% confidence intervals associated with the standardized beta weights, and then determine if the amount of overlap between the intersecting confidence intervals is less than 50%. If the amount of overlap is less than 50%, then you would have some justification to suggest that the difference in the standardized beta weight point-estimates is statistically significant ($p < .05$). The principle of such a suggested procedure is based on Cumming (2009) who demonstrated that 95% confidence intervals can be used to test the difference between a variety of different statistical point-estimates. In addition to the typical assumptions associated with multiple regression analysis, the procedure described here assumes that the confidence intervals are roughly the same size. Specifically, Cumming (2009) recommended that the larger confidence interval should be less than twice the size of the smaller confidence interval.

In order to obtain the appropriate 95% confidence intervals to apply the test described in this section, it is necessary to transform the independent and dependent variables into z-scores. Then, run the multiple regression analysis through the bootstrap utility with the request for 95% confidence selected. This is precisely the same procedure described in the

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bivariate regression chapter. The distinction in this example is that there are two beta weights for which 95% confidence intervals will be estimated. The confidence intervals can then be used to test the difference between the standardized beta-weight point-estimates. When I ran the analysis, I obtained the following result (**Watch Video 14.22: [Standardized Beta Weight Confidence Intervals in SPSS](#)**).

Coefficients ^a			
Model		95.0% Confidence Interval for B	
		Lower Bound	Upper Bound
1	(Constant)	-.117	.117
	Zscore(education)	.090	.375
	Zscore(IQ)	.031	.316

a. Dependent Variable: Zscore(earnings)

It can be seen that the education predictor standardized beta weight was associated with the 95% confidence intervals of .090 and .375. Similarly, the IQ predictor standardized beta weight 95% confidence intervals were .031 and .316. Note that, if you perform the analysis yourself, you may not get exactly the same results as reported above. The reason is that your 1000 bootstrapped re-samples are very unlikely to be exactly the same 1000 re-samples I obtained, when I performed in the analysis in SPSS. However, the results should be fairly consistent across re-analyses, particularly if a minimum of 2,000 re-samples is selected.

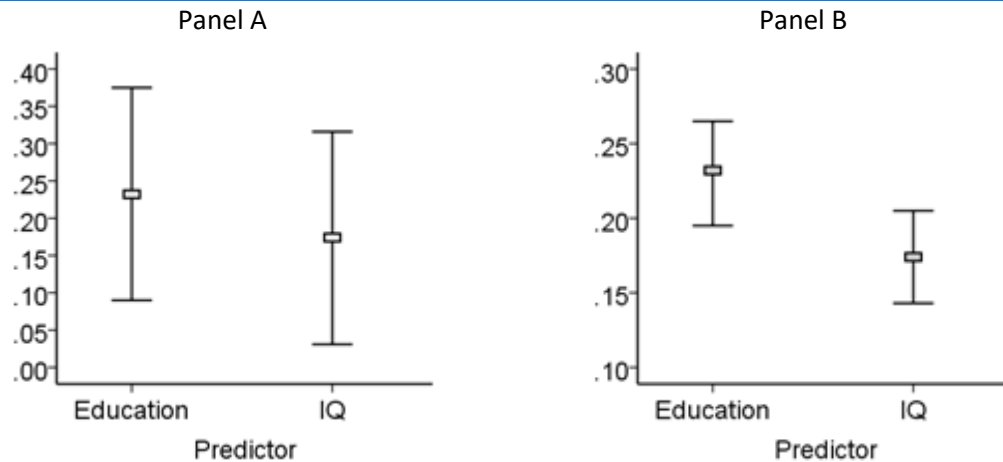
In order to display the results, if you wished to do so, the lower-bound, point-estimate, and upper-bound estimates can be displayed visually in a HILO chart in SPSS. I used the following syntax (**Data File: [dif between b](#)**) (**Watch Video 14.23: [Test the Difference Between two Beta Weights in SPSS](#)**).

GRAPH

/HILO(SIMPLE)=VALUE(UB LB Point).

The chart displayed in Figure 14.9 (Panel A) was produced by SPSS. Based on the visual inspection of the confidence intervals, it is clear that there was much too much overlap to suggest that there was a statistically significant difference between the standardized beta-weight point-estimates (.232 vs. .174). In fact, the confidence intervals intersect with the point-estimates, so, the p -value associated with the difference in the beta weights would be much greater than .05, in this case. Consequently, it may not be concluded that years of education was a more substantial contributor to the regression equation than IQ.

Figure C14.9. Standardized Beta Weights and Confidence Intervals



For the purposes of demonstration, I have created a fictitious set of data that were consistent with a statistically significant difference in the beta-weights, as the confidence intervals overlapped less than 50% ([Data File: dif between b sig](#)). The chart displayed in Figure C14.3 (Panel B) was created based on that fictitious data.

It can be seen that the lower-bound of education intersected with the upper-bound of IQ. However, the amount of overlapped was less than 50%; consequently, it may be suggested that the standardized beta weight point-estimates (.232 vs. .174) were statistically significantly different from each other ($p < .05$), in this supplementary fictitious case.

The exact calculations of the procedure involve two steps: (1) calculate the average length of the two overlapping arms; and (2) calculate the amount of overlap. If the amount of overlap is less than 50% of the length of the arms, reject the null hypothesis of equal standardized beta weights, $p < .05$. Thus, in this case, the average length of the overlapping arms (step 1) corresponded to $(.037 + .031) / 2 = .034$. The amount of overlap corresponded to $.205 - .195 = .010$. Thus, because .010 is less than 50% of the average arm length $(.010 / .034 = .29$ or 29%), the null hypothesis was rejected, $p < .05$. Finally, I'll note that the procedure described here assumes that the sample size is at least 30 (Cumming, 2009).

Koenker Test of Homoscedasticity

The Koenker test is described in more detail in the bivariate regression chapter (Chapter 12). The same SPSS syntax file described in the bivariate regression chapter can also be applied to the multiple regression context. As can be seen below, the Koenker test was significant statistically ($p = .025$), which implies that the assumption of homoscedasticity was not satisfied (**Watch Video 14.24: Breusch-Pagan and Koenker in Multiple Regression via SPSS**). The Koenker test result is very similar to the test of homoscedasticity result reported in the Foundations section of this chapter, based on the Pearson/Spearman correlation analysis

(i.e., correlation between the standardized predicted values and the absolute standardized residuals). As the assumption of homoscedasticity has not been satisfied, I cannot trust the standard errors and/or the p -values associated with the multiple regression beta-weights. Consequently, in this case, I need to try to deal with that somehow (e.g., conduct a Wild bootstrap multiple regression; adjust the standard errors; see remaining topics in this chapter). Hypothetically, had the preferred Koenker test not produced a statistically significant result (i.e., $p > .05$), I wouldn't have to do any extra work. Instead, I could potentially trust the non-adjusted standard errors and p -values outputted by SPSS (or any other program).

```
----- Breusch-Pagan and Koenker test statistics and sig-values -----
              LM              Sig
BP              83.246          .000
Koenker          7.382          .025

Null hypothesis: heteroskedasticity not present (homoskedasticity)

if sig-value less than 0.05, reject the null hypothesis
```

Dealing with Heteroscedasticity: Wild Bootstrap

An interesting question is whether bootstrapping assumes homoscedasticity. Based on the current literature, it would appear that bootstrapping does assume homoscedasticity, unless one uses a special type of bootstrapping known as the wild bootstrap (Wu, 1986; Flachaire, 2005). The bootstrapping performed by SPSS by default is not based on the wild bootstrap (neither the p -value nor the confidence intervals). Fortunately, it is possible to perform a wild bootstrap analysis in SPSS, however, one must use syntax to do so. I used the education, intelligence, and earnings data file for this example ([Data File: education earnings N 250](#)).

In order to conduct a wild bootstrap multiple regression analysis in SPSS, one must first conduct the multiple regression analysis (via normal theory is fine) and save the unstandardized residuals. SPSS will create a new variable named 'RES_1'. Next, perform the wild bootstrap analysis. Specifically, re-run the multiple regression analysis, however, this time, click on the Bootstrap button. Select the 'bias corrected accelerated' confidence intervals option. Click Continue. Click the 'Save' button and deselect the unstandardized residuals option. Click Continue. Do not click Ok. Instead, click on Paste, because you're going to have to modify the syntax in a minor way. When you click Paste, you should get the following syntax file:

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BOOTSTRAP

```
/SAMPLING METHOD=SIMPLE  
/VARIABLES TARGET=earnings INPUT= education IQ  
/CRITERIA CILEVEL=95 CITYPE=BCA NSAMPLES=2000  
/MISSING USERMISSING=EXCLUDE.
```

REGRESSION

```
/MISSING LISTWISE  
/CRITERIA=PIN(.05) POUT(.10)  
/NOORIGIN  
/DEPENDENT earnings  
/METHOD=ENTER education IQ.
```

You need to change the second line of syntax (i.e., `/SAMPLING METHOD=SIMPLE`) to `/SAMPLING METHOD=WILD(RESIDUALS=RES_1)`. When you run the above syntax, you should get the key multiple regression results in a SPSS table entitled ‘Bootstrap for Coefficients’:

Bootstrap for Coefficients

Model	B	Bootstrap ^a				
		Bias	Std. Error	Sig. (2-tailed)	BCa 95% Confidence Interval	
					Lower	Upper
1 (Constant)	-56261.421	11401.881	28168.197	.003	-110019.233	19829.764
education	3756.289	366.700	1105.860	.001	1624.326	7028.520
iq	501.978	-176.735	414.541	.086	-207.724	862.381

a. Unless otherwise noted, bootstrap results are based on 1000 wild bootstrap samples

It can be observed that, based on the wild bootstrap, IQ was not a statistically significant contributor to the model, $p = .086$. Furthermore, the lower and upper bound 95% confidence intervals were consistent with a non-significant p -value, as one value was negative (-207.72) and one value was positive (862.38). Thus, application of the wild bootstrap, at least in this case, overcame the awkward situation of a non-significant p -value, in conjunction with confidence intervals suggestive of a statistically significant effect, which was observed in the original bootstrap analysis above. (**Watch Video 14.25:** [Wild Bootstrap Multiple Regression in SPSS](#))

I’m not aware of any simulation research that has compared the wild bootstrap versus Hayes and Cai’s (2007) adjusted standard error technique for dealing with heteroscedasticity. My hunch, however, is that you will be steered in the right direction with the wild bootstrap, in most cases. Plus, you get to use the term ‘wild bootstrap’ in your report, which is pretty cool.

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A substantial amount of time and space was devoted to dealing with the problem of homoscedasticity in this chapter, because it is a serious problem, and one which I believe occurs fairly frequently with field data. However, researchers rarely test for it.

Dealing with Heteroscedasticity: Adjusted Standard Errors

Given that the assumption of homoscedasticity was not satisfied, one may be inclined to follow-up the analysis with an application of Hayes and Cai (2007) heteroscedasticity corrected standard errors SPSS macro, as described in the bivariate regression chapter. You can download the Hayes and Cai SPSS macro here: <https://tinyurl.com/ya5939sy> I obtained the following heteroscedastic corrected results (**Watch Video 14.26: [Heteroscedastic Adjusted Multiple Regression Standard Errors in SPSS](#)**):

```
Run MATRIX procedure:
```

```
HC Method
3
```

```
Criterion Variable
earnings
```

```
Model Fit:
```

R-sq	F	df1	df2	p
.1304	24.3258	2.0000	247.0000	.0000

```
Heteroskedasticity-Consistent Regression Results
```

	Coeff	SE(HC)	t	P> t
Constant	-56261.421	20170.0171	-2.7894	.0057
educatio	3756.2885	865.5699	4.3397	.0000
iq	501.9781	262.3053	1.9137	.0568

```
----- END MATRIX -----
```

It can be observed that education remained a statistically significant contributor to the equation ($b = 3756.29$, $t = 4.34$, $p < .001$), however, IQ was no longer a statistically significant contributor to the regression equation. Recall that in the normal theory estimation approach, IQ was associated with a standard error of 208.97 and a p -value of .017. However, based on the Hayes and Cai (2007) corrected approach, IQ's standard error was estimated to be 262.31. Thus, the normal theory estimation approach underestimated the IQ variable's standard error. When estimated more accurately with Hayes/Cai approach, IQ's unstandardized beta weight of $b = 501.98$ was found to be associated with a $t = 1.91$, $p = .0568$, which is not significant statistically. Again, if I had to choose, I would opt for the wild bootstrap results. However,

Hayes and Cai's (2007) method is a viable option, as well, when the assumption of homoscedasticity has been violated.

Multiple R^2 Increased but p -value Decreased?

Multiple R^2 is tested for statistical significance via an ANOVA table in many statistical programs. However, do not get mixed up here. A multiple R^2 is really just a "special" Pearson correlation coefficient (squared), not an ANOVA. As demonstrated above, multiple R was equal to the Pearson correlation between the predicted values of Y and the actual values of Y .

One phenomenon of multiple R^2 that often surprises people is that the p -value associated with a multiple R value can actually increase, even when the inclusion of an additional independent variable is associated with a statistically significant increase in multiple R . Although counter-intuitive, this phenomenon is entirely possible. Think of it this way. Suppose you had a really solid predictor of Y in a bivariate regression analysis which yielded a multiple R of .65. The bivariate regression analysis, in this case, will have a good level of confidence that the multiple R point estimate is somewhere around .65. However, suppose you added a really crappy predictor of Y to the regression equation (say, $b = .08$). Yes, the additional predictor might be a statistically significant contributor to the regression equation, but it's still relatively crappy. And, yes, the multiple R^2 value may increase in numerical value very slightly with the addition of the relatively crappy predictor to the model. However, the confidence interval around the multiple R will blow out as a consequence. Therefore, the p -value associated with the multiple R -value will increase, even if the multiple R value also increases. Essentially, when you add a relatively crappy predictor to a regression equation, you dilute the quality of the regression equation with respect to confidence in the overall prediction of the model. Most researchers accept the price of less confidence in the multiple R -value, if it means they have more statistically significant predictors to report and discuss, even if one or more of the predictors are relatively weak contributors to the model.

It is also worth noting that you can have a case where the multiple R -value is statistically significant, but none of the individual beta weights are statistically significant. Recall that the multiple R -value is based on a Pearson correlation between an optimally weighted variable (i.e., predicted Y values) and the actual Y values. The analysis used to determine whether any of the independent variables were statistically significant contributors to the regression equation on their own is distinct to the analysis relevant to the statistical significance of a multiple R . Typically, when there is a disconnect between the multiple R -value and the individual beta weights with respect to statistical significance, one or more of the beta weights were likely nearly statistically significant. In my experience, the inconsistency is always where the multiple R -value is statistically significant, but none of the beta weights are. I've never come across a multiple regression where the multiple R -value was not statistically significant, but one or more of the beta weights were.

Evaluating the Importance of Predictors: Relative Importance Analysis

Thus far, I have emphasized the interpretation of standardized-beta weights and semi-partial correlations, with respect to evaluating the importance of predictors in a multiple regression analysis. As far as evaluating the unique contributions of each predictor to a regression equation, I believe the semi-partial correlation is probably a person's best option. However, because the semi-partial correlation is an approximate representation of only unique variance contributed to a multiple regression equation, and, typically, multiple R^2 is not simply a function of the sum of the squared semi-partial correlations, something is lost by evaluating only the unique contributions of each predictor. Specifically, what is lost is the common variance contributed to the regression equation by the predictors. This can be considered a big deal, as there are cases where a predictor variable is found to be the weakest unique contributor to a regression equation (i.e., smallest semi-partial r^2), but the variable shares the most amount of variance with the regression equation predicting the dependent variable. Arguably, such a superficially viewed weakest contributor to the regression should be given further consideration, in a full evaluation of the multiple regression results.

Considering the above, one method to help evaluate the *overall* (not just unique) importance of predictors included in a multiple regression equation is to conduct a relative importance analysis, a.k.a., relative weights analysis (Johnson, 2000).⁹ Unfortunately, most popular statistical programs do not include the option to include a relative importance analysis as part of a multiple regression. Fortunately, however, Lorenzo-Seva, Ferrando, and Chico (2010) published SPSS syntax that can conduct a relative importance analysis (you can download it [here](#)). In practice, it doesn't really add much to conduct a relative importance analysis with just two predictors, as the amount of common variance contributed to the regression equation will be equal across both predictors. Thus, I have not applied a relative importance analysis to the education, intelligence, and earnings multiple regression example discussed, thus far. However, things become more interesting, when there are three or more predictors in the multiple regression.

For example, Zhao, Golde, and McCormick (2007) were interested in predicting graduate student satisfaction with thesis supervisors. To that effect, they had graduate students rate their perception of their supervisor's academic advice giving ability, their supervisor's personal touch, their supervisor's career development advice, and the student's perception that they were being treated like slave-labor by their supervisor. I have simulated some data to correspond closely to the results reported by Zhao et al. (2007), although I

⁹ There is also Budesco's (1993) dominance analysis to consider. However, the results of a dominance analysis and a relative importance analysis almost always agree perfectly (Budesco & Azen, 2004). Shapley's Value Regression also provides results very similar to a Johnson's relative weights analysis, but Shapley's method is more complicated to compute, and, so, is less popular.

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generated the data to have a sample size of 135, rather than 3,568, to make the results more interesting ([Data File: phd_student_satisfaction](#)). When I ran the relative importance analysis with the Lorenzo-Seva et al. (2010) syntax file (**Watch: Relative Weights Analysis**), I obtained the following results.

First, the program produced a correlation matrix. As can be seen below, the academic advice variable ('aca_advi') was associated with the largest zero-order Pearson correlation (i.e., $r = .681$) with student satisfaction ('satisfac'). However, the remaining three predictors were also non-negligibly correlated with the dependent variable, as well as inter-correlated with each other.

Correlations Between Dependent Variable and Original Predictors

	satisfac	aca_advi	personal	career_d	cheap_la
satisfac	1.000	.681	.598	.576	-.266
aca_advi	.681	1.000	.631	.675	-.167
personal	.598	.631	1.000	.584	-.365
career_d	.576	.675	.584	1.000	-.013
cheap_la	-.266	-.167	-.365	-.013	1.000

Next, the program reported the multiple R and multiple R^2 , with corresponding 95% confidence intervals (a really nice touch). It can be seen below that 53.5% of the variance in graduate student satisfaction with their supervisor was accounted for by the four predictors.

Multilple R. Point-Estimate and bootstrap 95% confidence interval

P. Estim	Lower	Upper
.731	.662	.802

R square. Point-Estimate and bootstrap 95% confidence interval (Lc

P. Estim	Lower	Upper
.535	.439	.642

After the unstandardized beta-weights (not included here), the program reported the standardized beta-weights and corresponding 95% confidence intervals. As can be seen below, the academic advice predictor variable was associated with the numerically largest standardized beta-weight ($\beta = .423$). The perceived cheap labor variable ('cheap_la') variable was the only one of the four that was not found to be a statistically significant contributor to the multiple regression equation, as the lower and upper bound 95% confidence intervals overlapped zero. However, as discussed further below, it would be a mistake to think that a supervisor could simply treat their students as cheap labor and think it wouldn't have a negative impact.

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Standardized regression coefficients (Beta) .			
	satisfac	Lower	Upper
aca_advi	.423	.231	.586
personal	.175	.012	.336
career_d	.187	.009	.352
cheap_la	-.129	-.256	.015

Next, the program reported the structure coefficients. In multiple regression, a structure coefficient represents the linear association between a predictor and the predicted values derived from a multiple regression equation. It may be regarded as a competitor to the relative importance analysis approach to evaluating the importance of predictors. As can be seen below, the academic advice variable correlated .93 with the regression equation function, which was the numerically largest correlate. Interestingly, despite the fact that the cheap labor variable was not found to be a statistically significant unique contributor to the regression equation (i.e., non-significant beta-weight), the cheap labor variable, nonetheless, correlated with the regression equation function in a statistically significant way (lower and upper-bound estimates were both negative). This means that treating graduate students as cheap labor might impact their perception of their supervisors in a negative way. However, the impact might be via its influence on another variable (e.g., personal touch?).

Structure coefficients. Point-Estimate and bootstrap 95%			
	satisfac	Lower	Upper
aca_advi	.931	.861	.968
personal	.818	.699	.910
career_d	.788	.657	.887
cheap_la	-.364	-.567	-.115

Finally, after a couple of preliminary relative importance analysis results (not included here), the program reported the key relative weights. As can be seen below, the academic advice variable was associated with a relative weight of 42.0%, which implies that of the 53.5% percentage of variance accounted for in graduate student satisfaction with their supervisor, academic advice produced 42.0% of that predicted variance. Next, personal touch and career development produced nearly identical amounts of the predicted variance ($\approx 25\%$). Finally, cheap labor accounted for 7.0% of the predicted variance, an amount of predicted variance that was statistically significant, as the lower-bound exceeded zero. The reason that cheap labor's standardized beta-weight was not significant was because it failed to contribute *unique* variance to the regression equation. However, the relative importance analysis suggests that cheap labor did contribute predictive variance that was common to one or more of the other predictors included in the model.

Relative contribution to Multiple R (reported as percentages).

	satisfac	Lower	Upper
aca_advi	42.0	29.4	52.4
personal	25.2	16.2	35.5
career_d	25.9	16.4	36.9
cheap_la	7.0	1.4	16.6

A relative importance analysis is similar to a commonality analysis, which I described in the moderated regression section of this chapter. A commonality analysis is more detailed, as it partitions the common predictive variance across all of the possible terms (i.e., combinations of predictor variables). By contrast, a relative importance analysis combines the unique and common variance explained into a single term, and it ensures that the sum of the relative weights does not exceed 100%. For further details on the various methods to interpret the importance of predictors in multiple regression, consult Nimon and Oswald (2013).

Suppressor Effects

One of the more difficult statistical concepts to grasp is suppression, as I suspect it is considered a counter-intuitive phenomenon for most people. Generally speaking, suppression is considered observed when the magnitude of the effect between X and Y increases, by adding one or more variables to the (usually regression) analysis. Thus, in one approach to the analysis, the beta-weight between X and Y increases, by adding one or more additional variables to the model. Such an effect is a bit baffling, as adding one or more variables to a regression-type analysis would be more naturally be expected to *reduce* the magnitude of the beta-weight between X and Y , consistent with partial or full mediation. Thus, in some sense, suppression may be considered the opposite of mediation. However, you should not consider suppression as a form of moderation, as there are no product terms in a suppression analysis. Ultimately, moderation, mediation, and suppression are three different types of ‘process analysis’ (Hayes, 2013).

In the most typical scenario (classical suppression), suppression occurs when there is small amount of shared variance between the independent variable (X) and the dependent variable (Y) (which may or not be significant statistically). Additionally, there is (typically) a moderate amount of shared variance between another variable (suppressor, S) and X . Finally, and this is the surprising bit, there is little to no shared variance between S and Y . This pattern of correlation coefficients tends to produce evidence for a suppressor effect. Essentially, by controlling statistically for the shared variance between S and X , the nature of X becomes ‘purified’ with respect to its association with Y . Consequently, by including S in the model, the magnitude of the association between X and Y increases, in comparison to the zero-order association between X and Y . At this point, it is probably best to consider an example.

It has been well-established that people can, to some degree, report their own level of intelligence accurately (Freund & Kasten, 2012). Typically, researchers report a correlation of about .30 between self-reported IQ and objective intelligence measures (task-based IQ). Gignac (2018) hypothesized that a construct known as self-deceptive enhancement (similar to narcissism) would correlate positively with self-reported IQ, but not with task-based IQ. Based on such a pattern of correlations, one would expect that the association between self-reported IQ and task-based IQ would increase, if the influence of self-deceptive enhancement on the self-reported IQ scores were taken into consideration. Stated alternatively, if self-reported IQ scores could be 'decontaminated' of self-deceptive enhancement, then the association between a decontaminated self-reported IQ variable and task-based IQ would be expected to be larger than the association between the original (contaminated) self-reported IQ scores and the task-based IQ scores.

To test the hypothesis, Gignac (2018) administered a self-report IQ questionnaire (SRIQ), a task-based IQ test (TBIQ), and a self-report measure of self-deceptive enhancement (SDE) to a sample of 253 young adults. Gignac (2018) found the following Pearson correlations: SRIQ and TBIQ $r = .16$, $p = .011$; SRIQ and SDE $r = .44$, $p < .001$; SDE and TBIQ $r = -.05$, $p = .428$. Thus, self-reported IQ was only relatively weakly associated with task-based IQ. However, because self-deceptive enhancement was fairly substantially correlated positively with self-reported IQ and very weakly with task-based IQ, there was the possibility of a suppressor situation.

It should be acknowledged that there are several different types of suppressor situations (Paulhus, Robins, Trzesniewski & Tracy, 2004). The Gignac (2018) suppressor situation is consistent with 'classical suppression', where X and Y are (usually) relatively weakly inter-related, X and S are inter-related positively, and Y and S are essentially or entirely unrelated.¹⁰ Thus, the Gignac (2018) study may be regarded as consistent with classical suppression. Classical suppression may be argued to be best tested by comparing the difference between the squared zero-order correlation between X and Y and the squared semi-partial correlation between X and Y , controlling for the effects of S on X (McGrath, 2010; Smith, Ager Jr & Williams, 1992; Velicer, 1978).

Recall that the zero-order Pearson correlation between self-reported IQ and task-based IQ was $r = .16$, thus $r^2 = .026$. In order to obtain the semi-partial correlation between self-reported IQ and task-based IQ in SPSS, I entered SRIQ and SDE as independent variables and TBIQ as a dependent variable in a multiple regression. Based on the output from SPSS, I obtained a semi-partial correlation of $r = .202$ (see column 'Part' in the SPSS table below

¹⁰ A classical suppression situation would also be observed where X and S are unrelated, but Y and S are related. The main issue is that either the independent variable or the dependent variable is decontaminated of a source of "nuisance" variance (nuisance at least with respect to the association between X and Y).

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entitled 'Coefficients'). Thus, the squared semi-partial correlation was equal to $r^2 = .041$. To my knowledge, a method to test statistically the difference between a squared zero-order correlation and a squared semi-partial correlation has not yet been established. I suspect the sample sizes required to detect a significant difference in the r -squared values with respectable power will be forbidding.

Fortunately, the results of Paunonen and Lebel's (2012) simulation work can offer guidance with respect to evaluating possible suppressor effects from the perspective of effect size. Specifically, based on the results of Paunonen and Lebel (2012), moderate and substantial suppressor situations may be regarded as consistent with an increase of 1% and 2% in predictive validity, respectively. Thus, in the Gignac (2018) SRIQ and TBIQ case, the difference in the semi-partial r^2 and the zero-order r^2 was $\Delta r^2 = .015$ (i.e., $.041 - .026 = .015$)¹¹, which corresponds to a 1.5% increase, i.e., consistent with somewhere between a moderate and substantial classical suppressor effect.

Coefficients^a

Model	Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Correlations		
	B	Std. Error	Beta			Zero-order	Partial	Part
1 (Constant)	16.233	1.848		8.785	.000			
SRIQ	1.857	.568	.225	3.270	.001	.160	.203	.202
SDE	-.318	.148	-.148	-2.153	.032	-.049	-.135	-.133

a. Dependent Variable: TBIQ

Cooperative Suppression

It should be noted that many researchers would test a classical suppression hypothesis via multiple regression, by regressing Y onto both X and S , with the expectation that the X beta-weight will be observed to be statistically significantly larger than the X and Y zero-order correlation. Based on both theoretical and statistical grounds, I do not believe this approach is entirely appropriate for testing classical suppression hypothesis. Instead, the multiple regression approach is optimal to test a suppression situation known as cooperative suppression (a.k.a., reciprocal suppression; Paulhus, 2004). The cooperative suppression situation is consistent with the following pattern of zero-order correlations: X and Y relatively weakly inter-related; S and Y relatively weakly inter-related; and X and S inter-related negatively. Thus, with respect to their associations with Y , both the X and S variables are mutually suppressed. Therefore, when both X and S are included in a multiple regression to predict Y , their beta-weights are observed to increase in magnitude, in comparison to their

¹¹ The results reported in Gignac (2018) are slightly different, as the analyses were conducted with maximum likelihood estimation in Amos with single-indicator latent variables; whereas, here, they have been conducted with ordinary least squares estimation in SPSS, based on simulated data generated to correspond closely to the results reported in Gignac (2018).

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respective zero-order correlations with Y . It should be noted that, in the context of cooperative suppression, it is not entirely appropriate to label one variable the independent variable (X) and one variable the suppressor variable (S). Instead, both variables are independent variables and both variables are suppressor variables.

To provide an example of cooperative suppression, consider an investigation by Krus and Hoehl (1994) who noted that there were substantial inter-nation differences in incarceration rates. At the extremes, the United States of America had 455 people per 100,000 incarcerated, whereas India had 34 people incarcerated per 100,000. Krus and Hoehl (1994) found that unequal distribution of national income (i.e., ratio of national income received by the top 10% to the bottom 20%) and family disintegration (i.e., divorce rates) were correlated positively with national incarceration rates. Importantly, the two independent variables were inter-correlated negatively. Such a pattern of correlations is consistent with the possibility of cooperative suppression. In my view, cooperative suppression hypotheses are best tested with multiple regression. In this example, it may be expected that a multiple regression with incarceration rates regressed onto unequal income distribution and family disintegration will reveal predictor variable standardized beta weights greater than their corresponding zero-order correlations. Such a pattern of results would be consistent with cooperative suppression.

To demonstrate the analyses associated with testing for cooperative suppression, I have simulated data to correspond closely to the results reported by Krus and Hoehl (1994) ([Data File: incarceration_rates](#)). Based on a Pearson correlation analysis, I obtained the results reported in the SPSS table entitled 'Correlations'.

Correlations

		incarceration_ rate	income_ ratio	divorce_ rate
incarceration_rate	Pearson Correlation	1	.549*	.464
	Sig. (2-tailed)		.027	.070
	N	16	16	16
income_ratio	Pearson Correlation	.549*	1	-.214
	Sig. (2-tailed)	.027		.427
	N	16	16	16
divorce_rate	Pearson Correlation	.464	-.214	1
	Sig. (2-tailed)	.070	.427	
	N	16	16	16

*. Correlation is significant at the 0.05 level (2-tailed).

It can be seen that incarceration rate correlated positively with both income ratio (income inequality) and divorce rate at $r = .549$ and $r = .464$, respectively. Thus, greater levels of income inequality and greater levels of divorce were associated with greater levels of incarceration rates. Furthermore, the income ratio variable correlated negatively with divorce rate at $r = -.214$. Consequently, I regressed the income ratio and the divorce rate independent variables

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onto incarceration rate and obtained the following key results reported in the SPSS table entitled 'Coefficients':

Coefficients^a

Model	Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Correlations		
	B	Std. Error	Beta			Zero-order	Partial	Part
1 (Constant)	-91.247	45.271		-2.016	.065			
income_ratio	17.233	4.218	.680	4.086	.001	.549	.750	.664
divorce_rate	54.121	14.768	.610	3.665	.003	.464	.713	.596

a. Dependent Variable: incarceration_rate

It can be seen that the income ratio variable was associated with a standardized beta-weight equal to .680, which was numerically larger than the zero-order Pearson correlation between income ratio and incarceration rate (i.e., .549). Such a result suggests the presence of a suppression situation. Furthermore, the divorce rate variable was associated with a standardized beta-weight equal to .610, which was numerically larger than the zero-order Pearson correlation between divorce rate and incarceration rate (i.e., .464). Given both beta-weights were numerically larger than the respective zero-order Pearson correlation, there was evidence to suggest cooperative suppression.

In practice, researchers do not test the difference between a standardized-beta weight and a zero-order correlation statistically. I suspect the reason is due to that fact that such a statistical test has not been well-established, yet. Instead, researchers simply interpret the larger standardized beta-weights than the corresponding zero-order correlations as evidence for cooperative suppression. Based on such a practice, it may be contended that income inequality and divorce rate suppress, in a mutual way, their respective statistical prediction of incarceration rate. Ideally, a relatively easy to implement statistical test will be established, in the near future, to test the numerical difference between a standardized beta-weight and a zero-order correlation.

In light of the above, it may be best to focus upon the difference between the independent variable squared semi-partial correlation and the corresponding squared zero-order correlation. In this example, income inequality was associated with a squared semi-partial correlation and a zero-order correlation of .441 and .301, respectively, which is equal to a difference of .140. A value of .140 is much greater than the .02 expectation for a large suppressor effect identified by Paunonen and Lebel's (2012) simulation work. Similarly, divorce rate was associated with a squared semi-partial correlation and a zero-order correlation of .355 and .215, respectively. Again, the difference of .140 would suggest a large suppressor effect.

In contrast to the example above, cooperative suppression can be observed when the two independent variables have differentially directed zero-order correlations (one positive

and one negative), but are inter-correlated positively (Paulhus, 2004). Such a suppression situation can also be found to be associated with larger standardized beta-weights than zero-order correlations. Finally, there is the suppression situation where X_1 and is relatively weakly, positively inter-related with Y (say, $r = .10$) and X_2 is fairly substantially, positively inter-related with Y (say, $r = .45$). Finally, X_1 and X_2 are also fairly substantially, positively inter-related with each other. Such a pattern of correlations will result in increased validity coefficients for both X_1 and X_2 ; however, X_1 is found to be *negatively* associated with Y , controlling for the effects of X_2 in a multiple regression. Such a suppression situation is known as negative (or net) suppression (Paulhus et al., 2004). My impression, is that instances of negative suppression are far less than common than classical and cooperative suppression. Overall, contrary to popular belief, my experience suggests that instances of suppressor situations are somewhat more commonly observed in behavioral science data. Consequently, it is important to keep suppression as a possibility in mind, when conducting analyses. Additionally, the possibility of suppression is a reason to not cease any hypothesized mediation analyses, simply because X and Y are not found to be associated with a statistically significant zero-order correlation. True, there is no possibility of mediation, when X and Y are not associated with a significant zero-order correlation. However, there is the possibility of a suppressor situation, which may be explicable on theoretical grounds.

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Practice Questions

1: Does spanking young children lead to more behavioral problems?

Barnes, Boutwell, Beaver, and Gibson (2013) obtained some impressive, longitudinal data relevant to spanking and behavioral problems in children. At 2 years of age, individual differences in the participants' self-regulation was measured. At 4-years of age, externalizing behaviors and number of spankings were measured. Finally, at 5-years of age, externalizing behaviors were measured again. Test the hypothesis that number of spankings a child received at age 4 contributed statistically significantly to a regression equation designed to predict externalizing behaviors at age 5. Include the following additional variables into the multiple regression equation, as well: self-regulation, spankings at age 4, and externalizing at age 4 ([Data File: spanking_externalizing](#)) (**Watch: Spanking & Externalizing**).

2: How well can overall job satisfaction be predicted?

What predicts job satisfaction? Is it the commonly discussed rate of pay? Gruenberg (1980) was interested to understand the nature of job satisfaction. He collected data on job satisfaction among 792 employees. Specifically, 792 employees responded to the following question: "All things considered, how satisfied would you say that you are with your job? Would you say that you are completely satisfied, pretty satisfied, not very satisfied, or not at all satisfied? Additionally, the employees responded to items relevant to their rate of pay, job security, the quality of the people they worked with, adequacy of workplace, freedom to plan work, chance to learn new things, and chance to use abilities. Conduct a multiple regression analysis to determine which of the variables contribute statistically significantly to a regression equation created to predict job satisfaction ([Data File: job satisfaction predictors](#)) (**Watch: Predicting Job Satisfaction**).

3: The role of optimism and your future health

Chopik, Kim, and Smith (2015) were interested in investigating whether individual differences in optimism can have an impact on a person's future health, controlling for several other variables. The participants started the study at age 51 (or older). In the first year of the study, the participants provided information relevant to their gender, age, and general health. They also completed an optimism scale (example item: "In uncertain times, I usually expect the best."). Two years later, the participants provided information about their general health and optimism, again. Conduct a hierarchical multiple regression with gender, age, general health1 (time 1), and general health2 (time 2) in block 1. Then, add optimism1 (time 1) and optimism2 (time 2) in block 2. Did the second block increase the multiple R^2 in a statistically significant? Were both optimism1 and optimism2 statistically significant contributors to the regression equation? ([Data File: optimism_health](#)) (**Watch: Optimism and Health**).

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4: Predicting serum cholesterol with a multiple regression equation

Bønaa, K. H., & Thelle, D. S. (1991). Association between blood pressure and serum lipids in a population. The Tromsø Study. *Circulation*, 83(4), 1305-1314.

Advanced Practice Questions

5: Absence frequency versus total number of days absent and manager impression.

What has a bigger impact on managers' impression of employee absenteeism, the frequency of employee absent periods, or the total number of days the employee is absent? Breaugh (1981) collected official human resources data on absenteeism from 112 employees. Absenteeism was measured in two ways: (1) total number of days absent; and (2) total number of periods absent. Thus, a person may be absent for only one period within a year, but it could be for several days in a row. By contrast, a person may miss 5 days of work throughout a year, but each of those days were across the year (5 separate periods). Breaugh (1981) also collected data from each employee's manager with respect to their rating (impression) of work attendance. Consequently, these data can help answer the question of which type of absenteeism impacts a manager's impression more: total number of absent days versus total number of absent periods? Run a multiple regression with manager ratings as the dependent variable. Total number of days and total number of periods should be included as the independent variables. Test the difference between the two standardized beta-weights for statistical significance ([Data File: absence frequency total](#)) (**Watch: Absence Frequency**).

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